Name:____________

Double Award Science – Biology Unit 1 Foundation Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions your struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /60	Grade	Good Questions	Bad Questions	Reasons
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Score

Year	G	F	E	D	C	Max UMS
2017	11	14	17	20	24	28
2018	8	12	16	20	24	28
2019	6	10	14	19	24	29
2022	11	15	19	23	28	33
2023	9	13	17	21	26	31

Surname	Centre Number	Candidate Number
Other Names		0

GCSE – NEW

3430U10-1

S17-3430U10-1

SCIENCE (Double Award)

Unit 1: BIOLOGY 1
FOUNDATION TIER

WEDNESDAY, 14 JUNE 2017 – MORNING

1 hour 15 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**5**	
2.	**9**	
3.	**11**	
4.	**6**	
5.	**14**	
6.	**5**	
7.	**10**	
Total	**60**	

3430U101
01

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet.

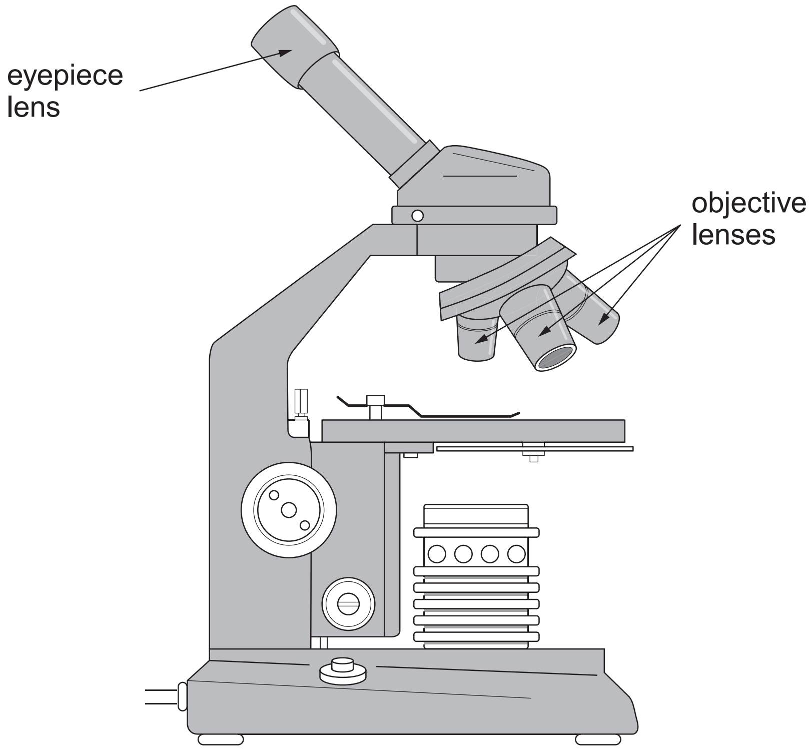
INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **5**(*c*) is a quality of extended response (QER) question where your writing skills will be assessed.

Examiner only

Answer **all** *questions.*

1. Rhys studies some plant tissue using the instrument shown below.



(a) State the name of the instrument shown in the diagram. [1]

....

(b) The table shows the magnification of each of the four lenses.

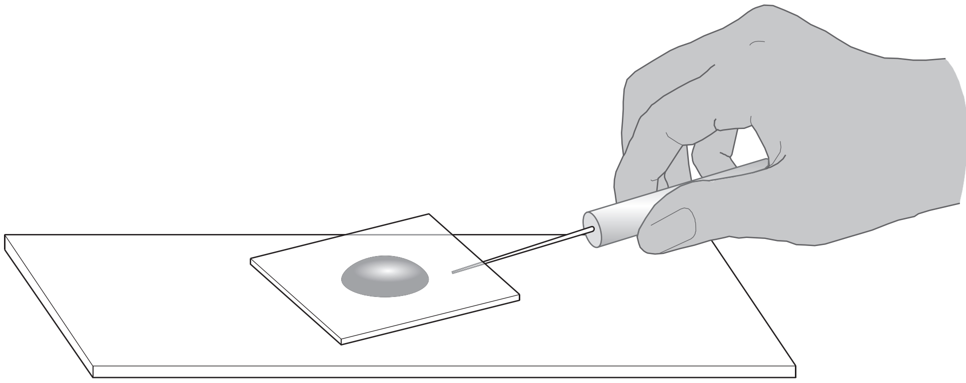
Lens	Magnification
eyepiece lens	×10
low power objective lens	×4
medium power objective lens	×10
high power objective lens	×40

Calculate the maximum magnification that is possible with this instrument. [2]

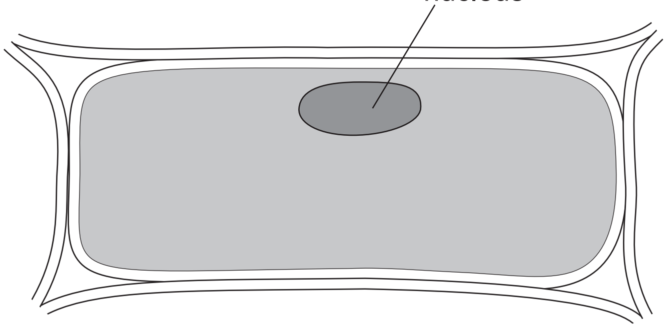
maximum magnification = × ....

Examiner only

Rhys places some of the plant tissue in water on a slide and lowers a cover slip on top as shown below.



He draws one cell from the tissue as seen under the maximum magnification.
His drawing is shown below.



(c) State what Rhys could have done to the plant tissue to show more detail of the cell structures. [1]

....

(d) State the function of the nucleus. [1]

....

....

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03

5

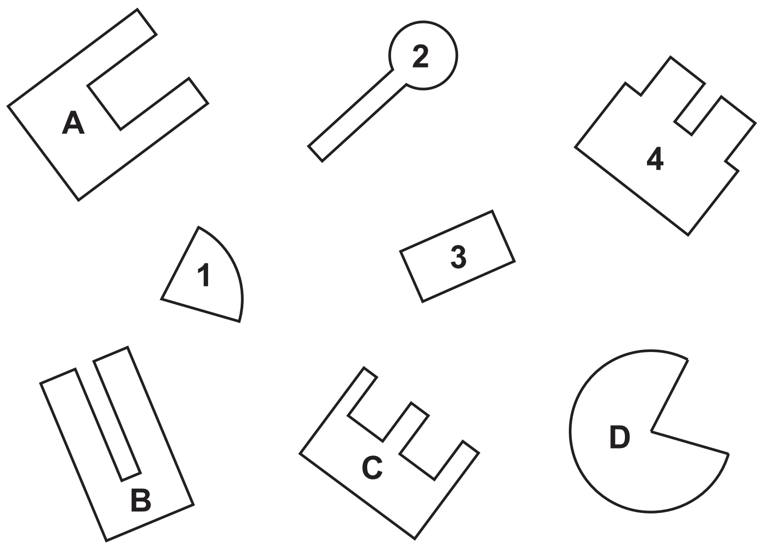
Examiner only

2. *(a)* Use some of the following words to complete the table about enzymes. [3]

fatty acids **lipids** **amino acids** **glucose** **glycerol**

Enzyme	Substrate	Products
protease	protein	....
lipase	....	.... and ....

(b) The diagram shows four enzymes **A – D** and four substrates **1 – 4**.

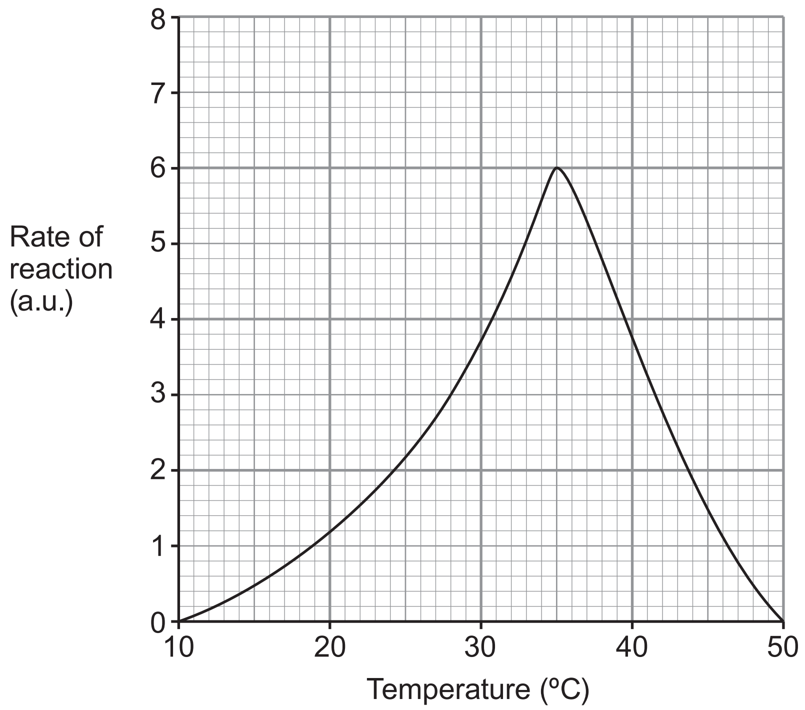


Use your knowledge of the lock and key theory to complete the table below by matching each enzyme to its substrate. [1]

Enzyme	Substrate
A	....
B	....
C	....
D	....

Examiner only

(c) The graph shows the effect of temperature on the rate of an enzyme controlled reaction between 10 °C and 50 °C.



(i) From the graph, describe the effect of temperature on the rate of the reaction between 10 °C and 50 °C. [3]

....

....

....

....

....

(ii) Most enzymes are denatured by boiling.
Use your answer to part *(b)* to help explain why a denatured enzyme can no longer work. [2]

....

....

....

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05

9

3. In a study of physical fitness levels, experts analysed a survey of adults in which they were asked how much they had exercised in the previous four weeks.

The report of the survey, which involved 1615 people, claimed that 30% of the British population did no exercise.

Reported in DailyTelegraph 05/06/15

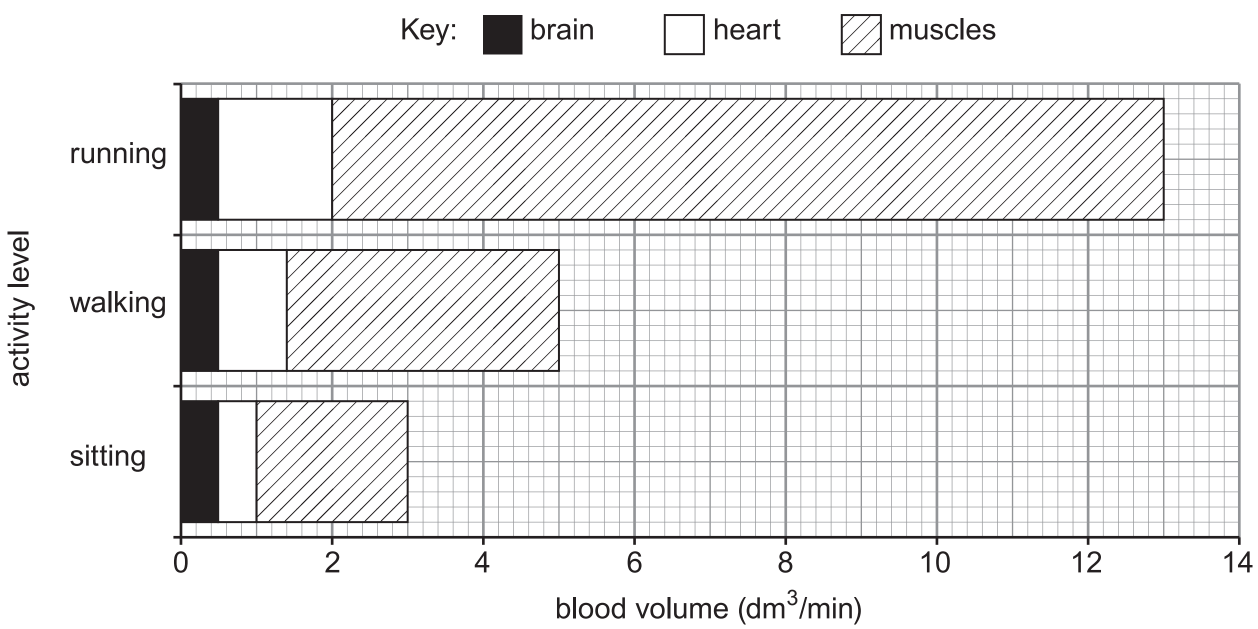
(a) Using the above information, calculate the number of people in the survey who did no exercise. [2]

Number of people = ...

(b) Why was it important to use a large number of people in this survey? [1]

....

(c) The bar chart shows the blood volume (dm^3/min) supplied to some parts of the body at rest and during exercise.



Examiner only

(i) Calculate the blood volume supplied to the muscles when sitting. [1]

Blood volume = ... dm^3/min

(ii) Calculate the increase in blood volume supplied to the muscles when running compared to sitting. [1]

Increase in blood volume = ... dm^3/min

(iii) Complete the following sentence. [1]

As blood flows through muscles, substances pass out of narrow blood vessels.

These blood vessels are called ....

(iv) Three of the following statements correctly describe the result of **an increase in blood volume** supplied to muscles.

Underline the **three** correct statements. [3]

More oxygen is supplied

More lactic acid is produced

More glucose is supplied

More anaerobic respiration takes place

More aerobic respiration takes place

(d) Explain why lack of exercise can result in obesity. [2]

....

....

....

....

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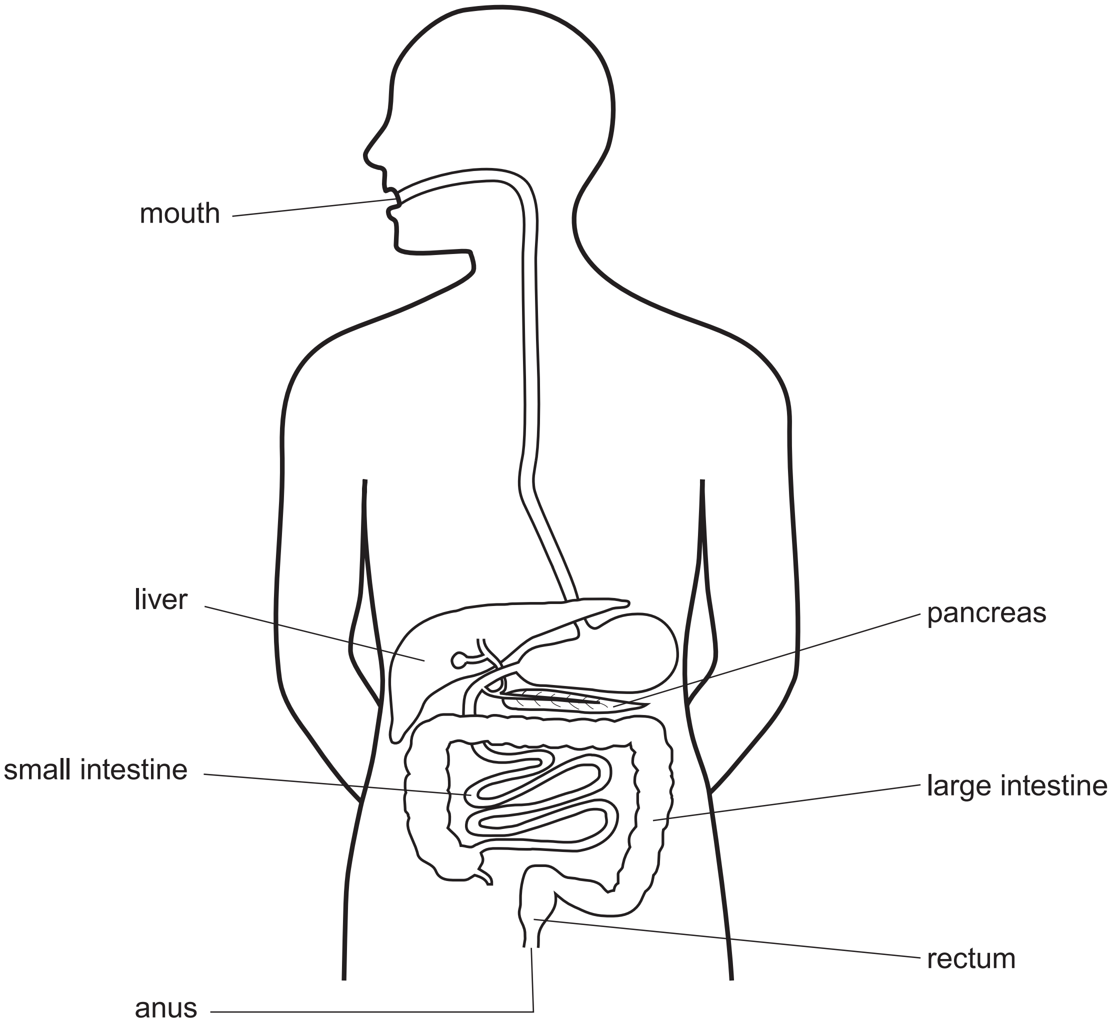
07

11

Examiner only

4. The diagram shows the human digestive system.



(a) State which of the **labelled parts** in the above diagram absorbs digested food molecules. [1]

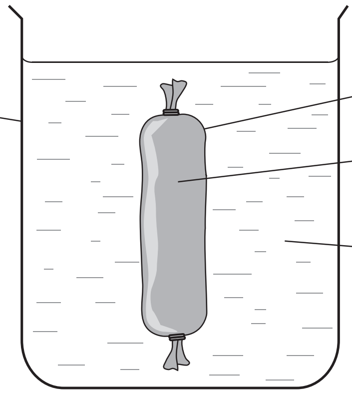
....

Examiner only

(b) Visking tubing acts as a model of absorption in the digestive system.

The diagram shows a Visking tubing bag filled with a solution of starch and glucose, in a beaker of water.

Samples from inside the Visking tubing bag and the water in the beaker were tested for starch and glucose at the start. This was repeated after 10 and 20 minutes.



(i) **Complete the table below** by writing a ✓ or x in each space to show the expected results at 10 and 20 minutes. [3]

key: ✓ molecule present x molecule absent

Sample tested	Tested for	Start	Time after start (minutes)	
			10	20
Contents of Visking tubing bag	starch	✓	✓	✓
	glucose	✓	✓	....
Water in the beaker	starch	x	....	....
	glucose	x	....	....

(ii) Use your knowledge of the size of starch and glucose molecules to explain the expected results **at 20 minutes** for **the water in the beaker**. [2]

....

....

....

....

....

6

3430U101 09

Examiner only

5. Smoking is the largest single cause of avoidable ill health and early death in Wales. In order to reduce this, the Welsh Government has focused on three areas in recent years: discouraging young people from starting to smoke, helping smokers to give up and extending smoke-free environments.

(a) Tobacco smoke contains carcinogens.

State the type of disease caused by carcinogens. [1]

....

(b) The table below shows the percentage of the adult population of Wales between 2007 and 2014 who smoked cigarettes.

Year	Percentage of the adult population who smoke (%)
2007	24
2008	24
2009	23
2010	22
2011	22
2012	21
2013	20
2014	19

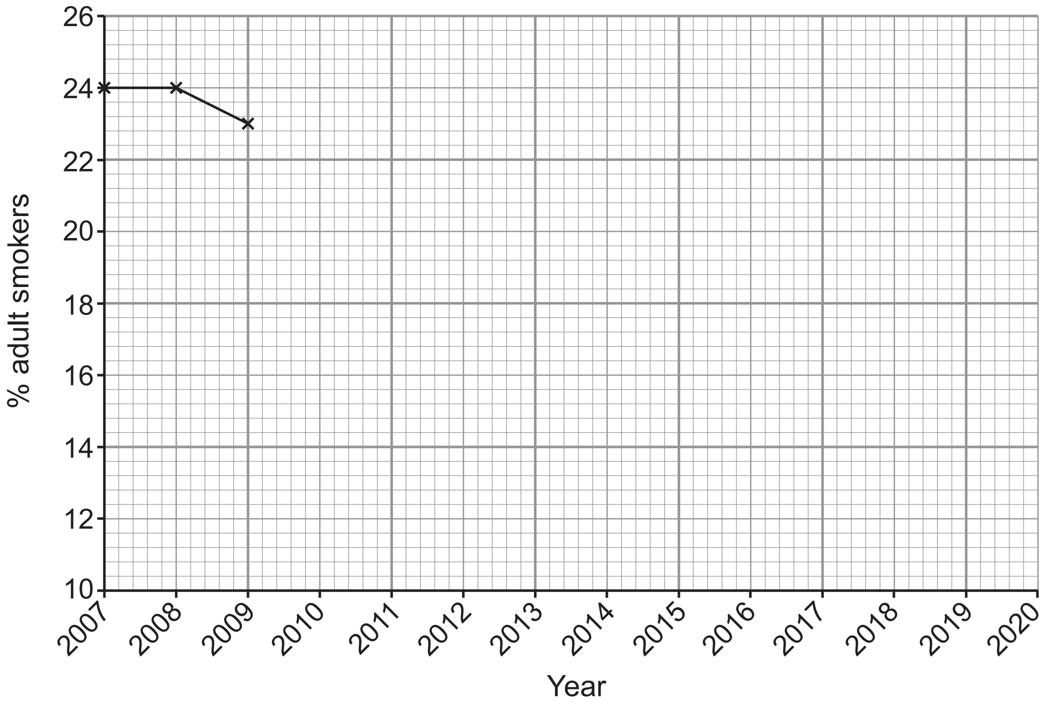
Sources: National Public Health Service for Wales, Welsh Assembly Government, Office for National Statistics

(i) Complete the graph of the data above on the grid opposite by: [3]

I. plotting the points from 2010 to 2014

II. joining all the plots using a ruler.

Examiner only



(ii) In 2007, the Welsh Government banned smoking in enclosed public places as part of a policy to reduce the percentage of adults who smoke.

Give the evidence that the ban had no immediate effect on the percentage of adults who smoke. [1]

....

(iii) In 2012, the Welsh Government produced a Tobacco Control Action Plan. One aim was that by 2020 no more than 16% of adults would smoke.

I. **Extend the line on the graph to predict** the % of adult smokers in 2020. Write your answer below. [1]

adult smokers in 2020 = ... %

II. Your prediction in part (I) is based on an assumption. What does your prediction assume? [1]

....

....

(iv) Suggest **one** *other* step that could further reduce the percentage of adults who smoke. [1]

....

....

Examiner only

(c) The breathing rate is the number of times a person breathes in and out in one minute. You have 100 volunteers. Of these, 50 are cigarette smokers and 50 do not smoke.

Plan an investigation to answer the following question:

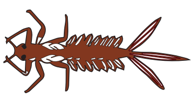
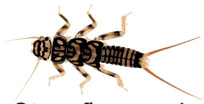
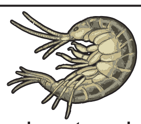
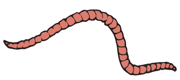
'Is the breathing rate of smokers different from that of non-smokers?'

Include at least **two** factors which should be kept the same in order to make it a fair test. [6 QER]

14

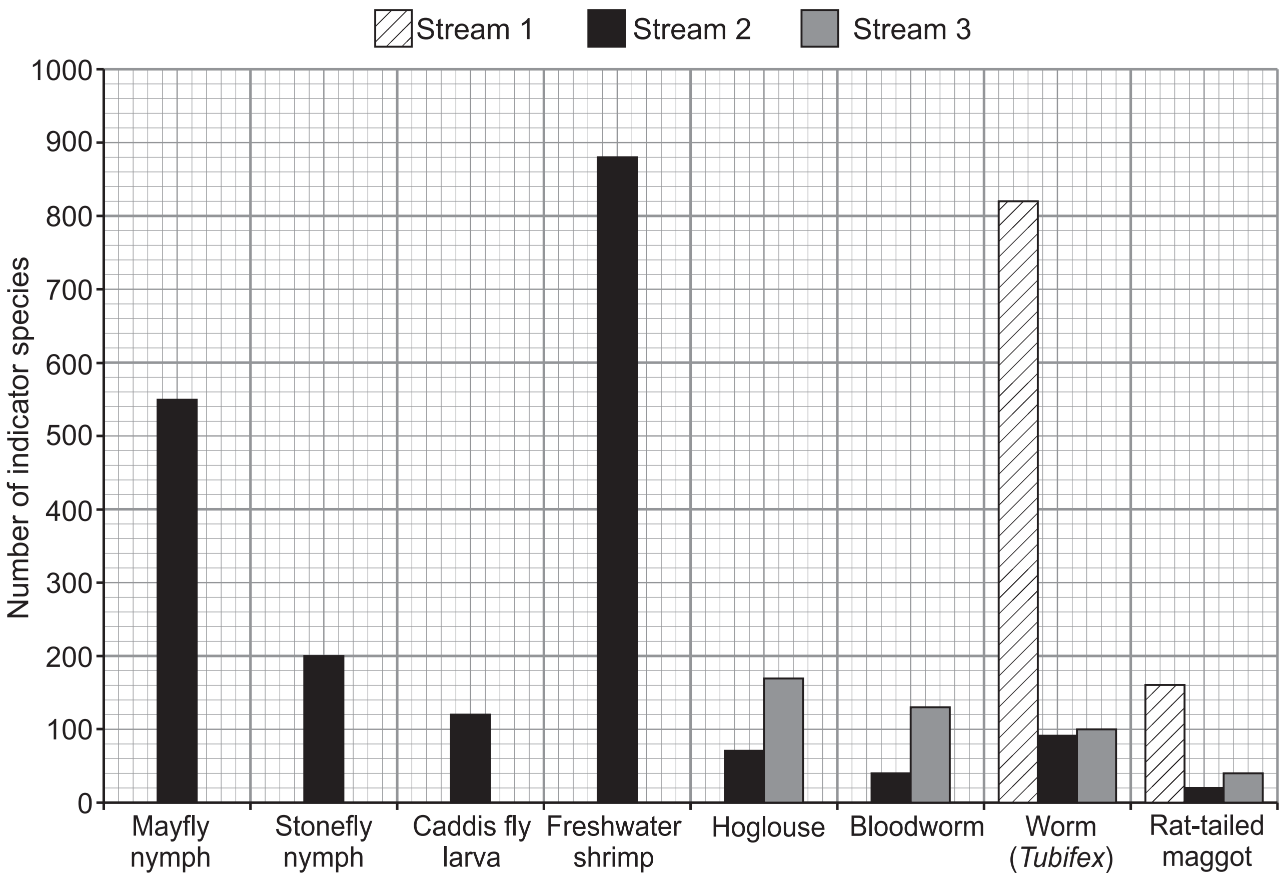
6.

Following flooding in parts of Wales in 2015, environmental scientists investigated nitrate pollution in three different streams. They counted the numbers of eight different indicator species in equal lengths of the three streams, on the same day. The indicator species they counted are shown in the chart below.

GROUP	INDICATOR SPECIES		POLLUTION LEVEL
A	 Mayfly nymph	 Stonefly nymph	If you find these animals in your stream, then there is **LITTLE OR NO POLLUTION**
B	 Caddis fly larva	 Freshwater shrimp	If you find these animals, but none from Group **A**, then there may be **SLIGHT POLLUTION**
C	 Hoglouse	 Bloodworm	If you find these animals, but none from Groups **A** or **B**, then there is probably **MEDIUM POLLUTION**
D	 Worm (*Tubifex*)	 Rat-tailed maggot	If you find these animals, but none from Groups **A**, **B** or **C**, then there is **HIGH POLLUTION**
E	No live animals found		If you find no animals at all, then there is **VERY HIGH POLLUTION**

Examiner only

Graph showing the number of indicator species found in each of the streams 1, 2 and 3



(a) (i) Using the graph above and the chart on the opposite page state the pollution level in each of the three streams. [2]

stream **1** ....

stream **2** ....

stream **3** ....

(ii) Suggest **two** possible **sources** of nitrate pollution in streams. [2]

....

....

(b) Name **one** type of indicator species which could be used to assess levels of air pollution. [1]

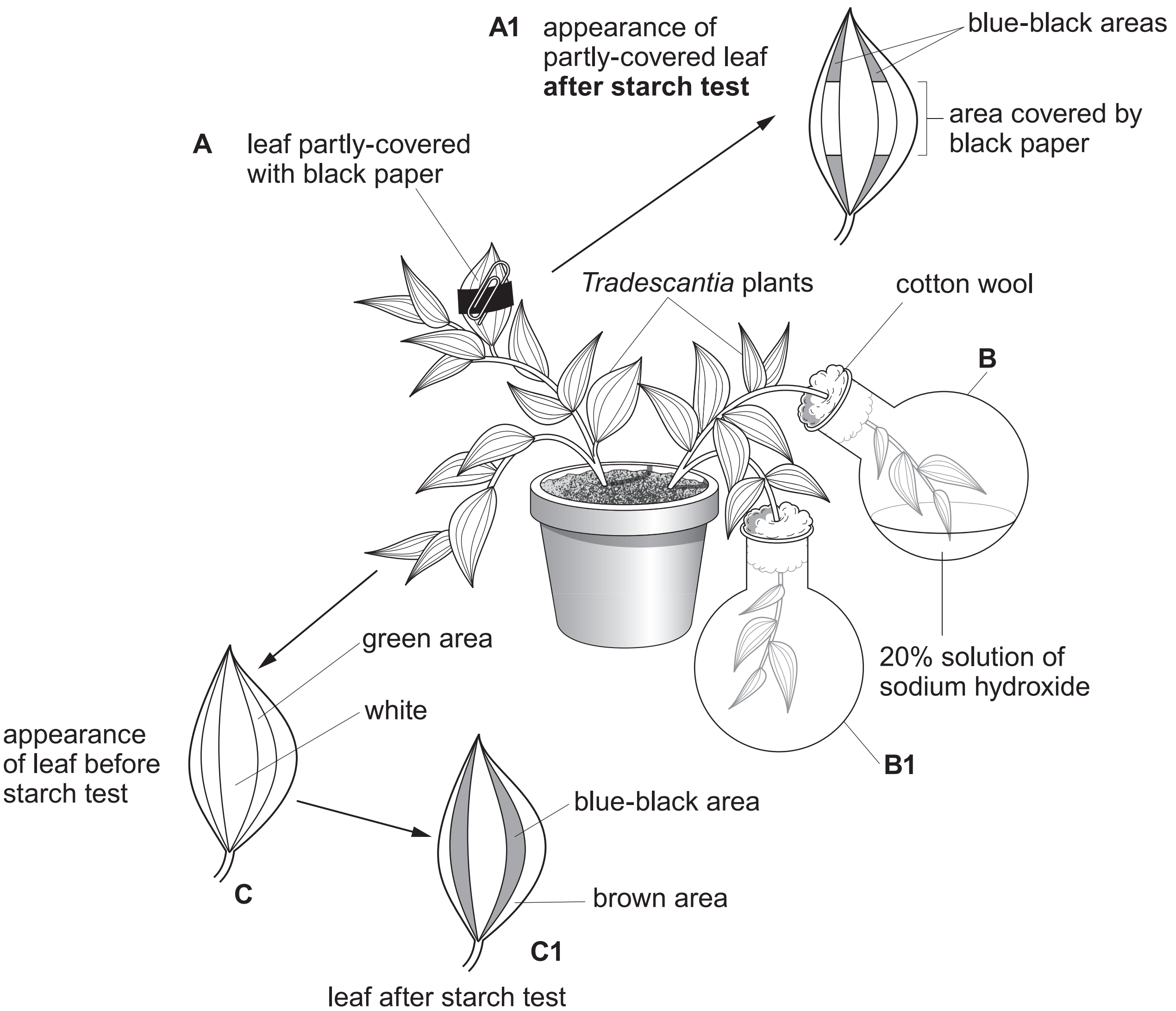
....

5

7. *Tradescantia* is a plant whose leaves have green and white areas.

Tradescantia leaves

In the experiment below, a *Tradescantia* plant is used in an investigation to demonstrate **three** factors, **A**, **B** and **C**, needed for photosynthesis.



Examiner only

(a) (i) State the factor needed for photosynthesis, which is being demonstrated in each of the following: [3]

A ....

B ....

C ....

(ii) How could the experiment in flasks **B** and **B1** be improved? Explain your answer. [2]

....

....

(iii) Before the apparatus was set up, the *Tradescantia* plant was kept in a dark cupboard for 48 hours. Explain the reason for this. [2]

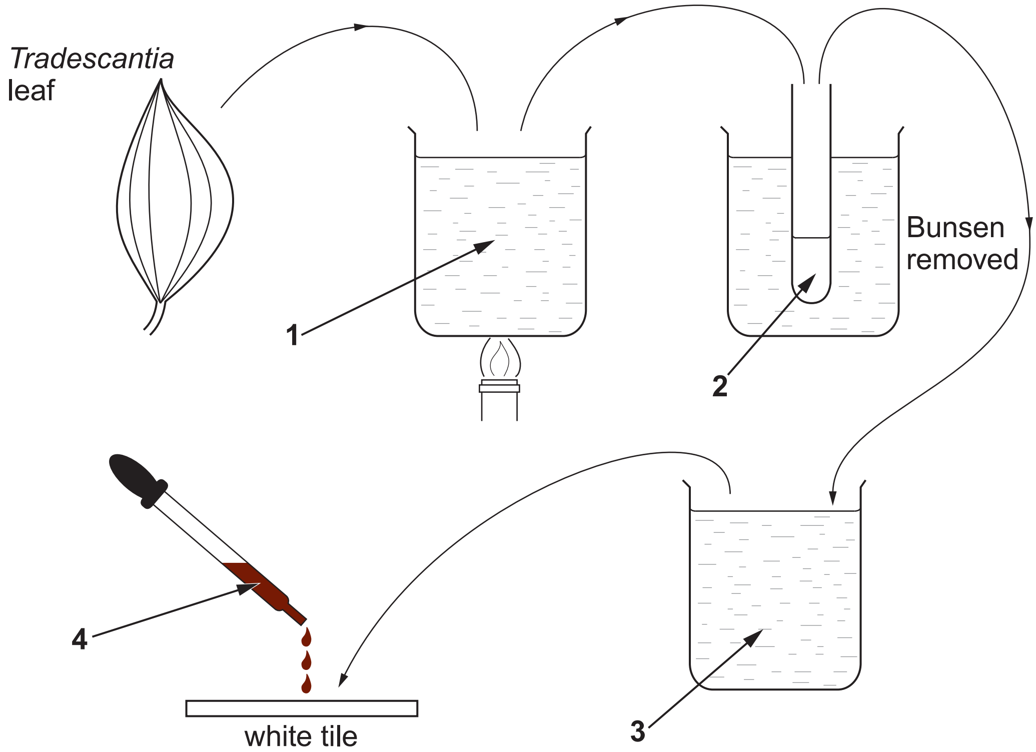
....

....

....

Examiner only

(b) The diagram below shows the stages in testing a leaf for starch.



Name the liquids **1**, **2**, **3 and 4** shown in the diagram. [3]

1 ....

2 ....

3 ....

4 ....

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END OF PAPER

Surname	Centre Number	Candidate Number
Other Names		0

GCSE – NEW

3430U10-1

S18-3430U10-1

SCIENCE (Double Award)

Unit 1: BIOLOGY 1
FOUNDATION TIER

MONDAY, 11 JUNE 2018 – MORNING

1 hour 15 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**7**	
2.	**6**	
3.	**8**	
4.	**9**	
5.	**8**	
6.	**7**	
7.	**6**	
8.	**9**	
Total	**60**	

3430U101
01

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

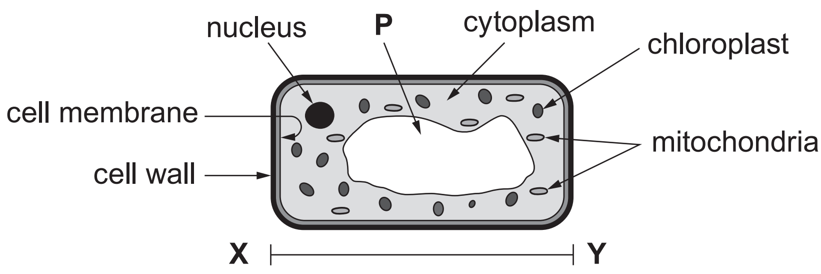
The number of marks is given in brackets at the end of each question or part-question.

Question **6**(*a*) is a quality of extended response (QER) question where your writing skills will be assessed.

Examiner only

Answer **all** *questions.*

1. The diagram shows a plant cell. Some structures have been labelled.



(a) (i) Use a ruler to measure the length of the cell at **X – Y** in mm. [1]

length at **X – Y** = ... mm

(ii) The diagram is magnified ×400.

Use your answer to part (i) to calculate the actual length of the cell. [1]

actual length = ... mm

(b) State the name of structure **P**. [1]

....

Examiner only

(c) Complete the following table about plant cells. [4]

Name of structure	Function
....	respiration
....	controls entry and exit of materials
chloroplasts	....
....	contains chromosomes

7

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03

Examiner only

2. *(a)* Complete the following sentences about enzymes using some of the words from the list below. [3]

digestion **photosynthesis** **diffusion** **respiration** **osmosis**

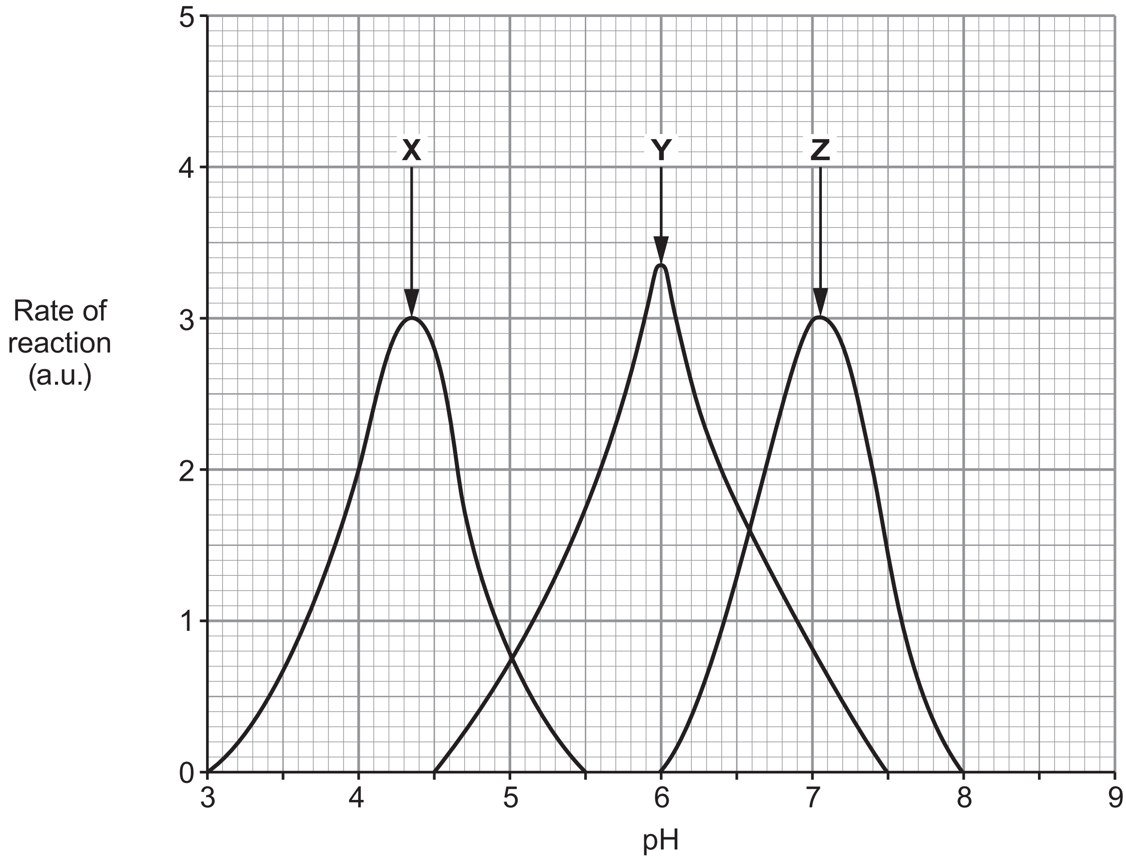
Some enzymes break down large molecules into small molecules, for example during

.... and ....

Other enzymes build up large molecules from small molecules, for example during

....

(b) The graph shows the activity of three enzymes, **X**, **Y** and **Z** at different pH values.



(i) State the enzyme (**X**, **Y** or **Z**) that is active only in acid conditions. [1]

....

Examiner only

(ii) Underline the pH range at which **both** enzymes **Y** and **Z** work. [1]

Between pH 4.5 and pH 8.0

Between pH 4.5 and pH 7.5

Between pH 6.0 and pH 7.5

(iii) Enzyme **Z** is found in saliva.

State a structure in the body that produces saliva. [1]

....

6

3430U101 05

Examiner only

3. The photograph shows an insect called an aphid *(Aphis)*.

Aphids damage crops such as barley, by making holes in the leaves. They then suck out sugar solution through the holes. A thick layer of fungi can then grow on the damaged leaves, so they absorb less light energy.

Farmers may use pesticides on their crops. Pesticides are effective, but may also be toxic to harmless organisms. Another approach is to release insects such as ladybirds onto the crop. Ladybirds are secondary consumers that are common in many food chains. They target pests such as aphids and so reduce their numbers.

(a) Using **only** the information above, give the evidence that:

(i) barley is photosynthetic; [1]

....

(ii) aphids are primary consumers; [1]

....

(iii) ladybirds are carnivores; [1]

....

(iv) using ladybirds is less likely to damage the environment than using pesticides. [2]

....

....

....

Examiner only

(b) (i) A farmer growing barley aims to harvest 8.0 tonnes/hectare.
Aphids reduce her harvest by 15 %.

Calculate the loss due to the aphids in tonnes/hectare. [2]

loss = ... tonnes/hectare

(ii) The farmer is paid £117.00 per tonne for her barley.

Use your answer to (i) to calculate how much money the farmer loses per hectare due to aphid damage. [1]

loss = £.................................... per hectare

3430U101
07

8

Examiner only

4. An investigation compared the composition of inspired and expired air. This is shown in the table below.

Gas	% Concentration of air	
	inspired	expired
oxygen	20.9	16.0
carbon dioxide	0.04	4.0
water vapour	variable	variable
nitrogen	78.1	78.1

(a) (i) Calculate the difference in the % concentration of oxygen in the expired air and the inspired air. [1]

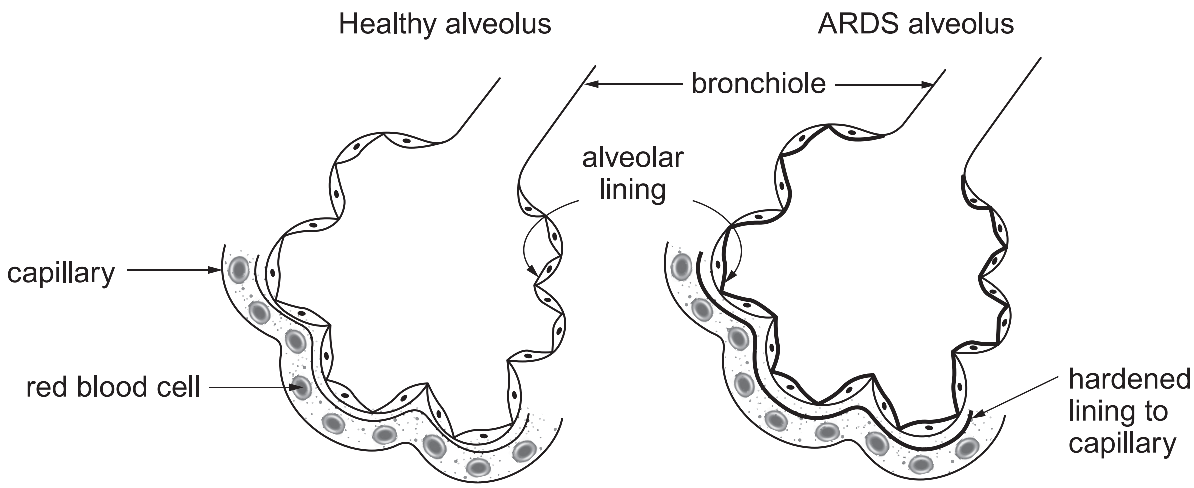
difference = ... %

(ii) State the process in cells that uses oxygen and glucose to release energy. [1]

....

(b) People with a disease called ARDS (Acute Respiratory Distress Syndrome) have difficulty getting enough oxygen.

The diagrams show a healthy alveolus and an alveolus from someone with ARDS.



(i) **Draw an arrow on the diagram** of the **healthy** alveolus, to show the direction of movement of oxygen through the alveolar lining. [1]

Examiner only

(ii) Describe **two** differences you can see between the two diagrams and explain why people with ARDS have difficulty getting enough oxygen from inspired air. [3]

....

....

....

....

....

(c) Many different types of molecules pass across cell membranes.

Complete the table below to show the direction of movement of molecules between blood and muscles. Place **one** tick (✓) in each row. [3]

Molecule	**From** blood **to** muscles	**From** muscles **to** blood	**To and from** blood and muscles
oxygen			
carbon dioxide			
water			

3430U101 09

9

Examiner only

5. *(a)* Give **one** harmful effect that may result from each of the following lifestyle choices. [2]

(i) eating excess salt (sodium chloride)

....

(ii) smoking cigarettes

....

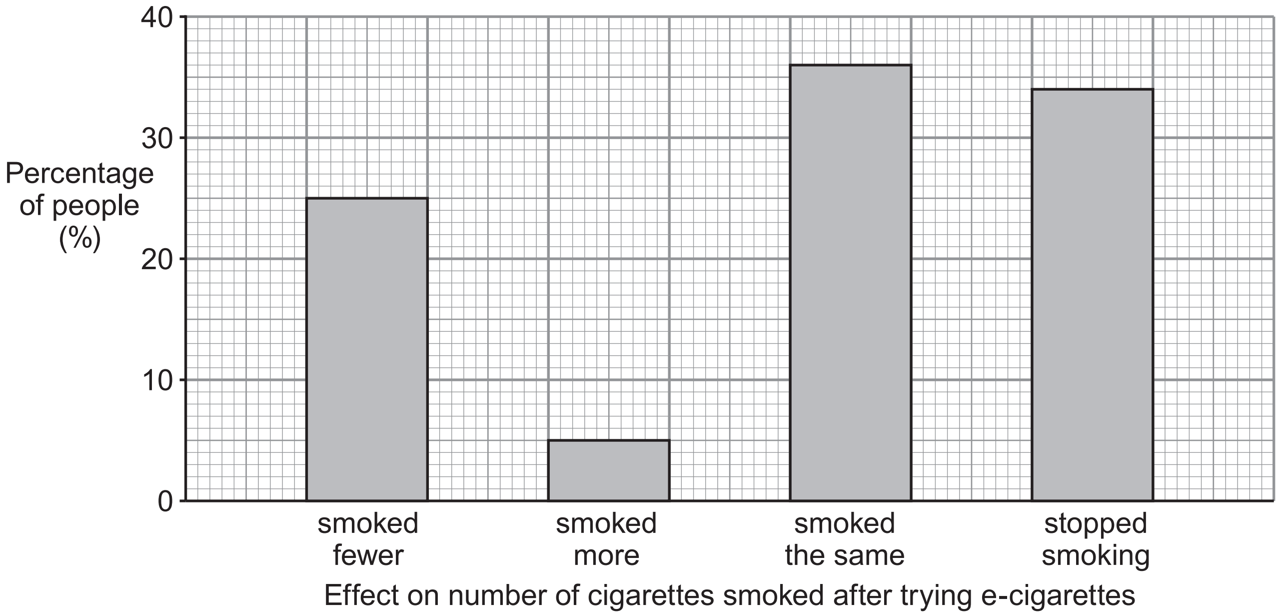
(b) Many people who smoke cigarettes would like to stop.

One method aimed at helping people stop smoking is by using an e-cigarette as shown in the photograph.

In 2016, 838 young people living in Wales were asked about using an e-cigarette.

Some of those questioned were cigarette smokers who had tried using e-cigarettes to help them cut down or stop smoking tobacco. *[ASH survey 2016]*

The bar chart shows how trying e-cigarettes affected their cigarette smoking.



Examiner only

(i) How successful was using e-cigarettes in helping young people cut down or stop smoking? Use **all** the results in the bar chart to justify your answer. [4]

....

....

....

....

....

....

....

....

(ii) The original sample involved 838 young people. A second survey is planned to include 10 000 people. Suggest **two** factors which should be considered in the selection of the people to take part in the second survey, to make it more representative of the population of Wales. [2]

....

....

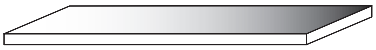
....

....

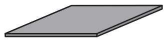
8

Examiner only

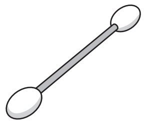
6. *(a)* Describe the method you would use to make a slide of your own cheek cells using the apparatus below. [6 QER]



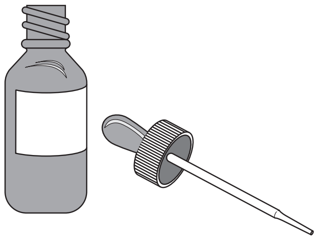
glass slide



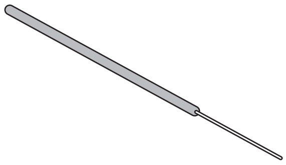
cover slip



cotton wool bud



methylene blue stain



mounted needle

Examiner only

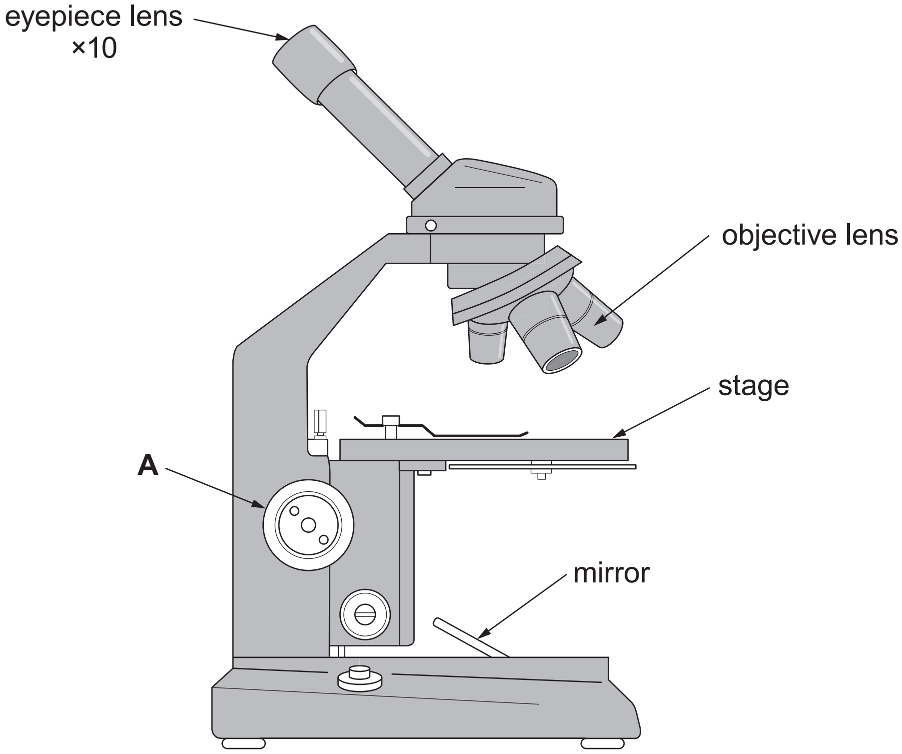
(b) Make a large drawing of one cheek cell in the box below.

Include the cell structures you would see when viewed at the highest magnification of a light microscope. No labels are required. [1]

7

Examiner only

7. Rhys was asked by his teacher to set up a light microscope so that he could view some cells at a magnification of ×100. The microscope had three objective lenses of ×4, ×10 and ×40 magnifications. Rhys was also given a prepared slide of muscle cells.



(a) Explain how Rhys could view the muscle cells at a magnification of ×100. [2]

....

....

....

(b) State the function of structure **A** on the diagram. [1]

....

Examiner only

(c) When Rhys viewed the muscle cell under the microscope he could see that the cells were not found on their own, but were grouped together in large numbers.

(i) Muscle cells are described as being specialised cells. State the advantage to the organism of having specialised cells. [1]

....

....

(ii) State the name given to a large number of the same cells grouped together. [1]

....

(d) In biology, what is meant by the term organ? [1]

....

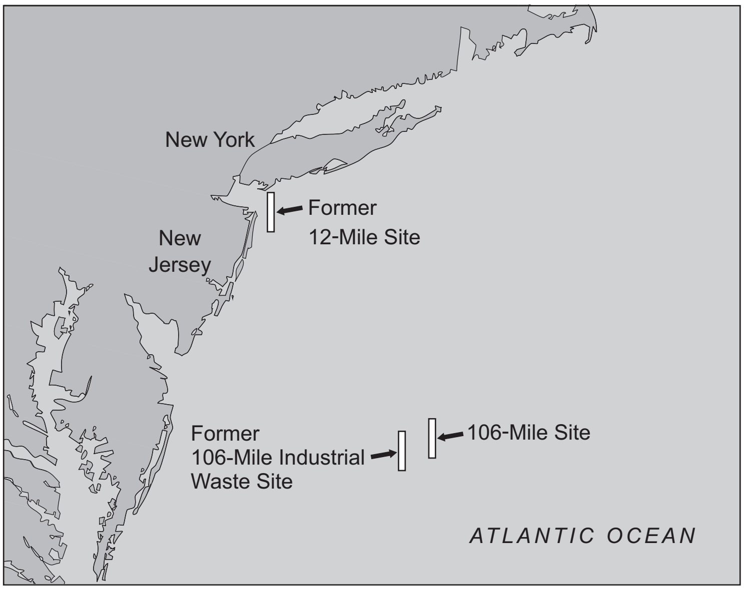
....

6

8. In the past, many countries, including the UK, have disposed of sewage sludge in the open ocean. A famous example of this practice is the '106 mile' dump site in the North West Atlantic. This site, 106 miles off the east coast of the USA, served the populations of New York and New Jersey. Prior to the use of the '106 mile' dump site sewage sludge was disposed of at the '12 mile' dump site.

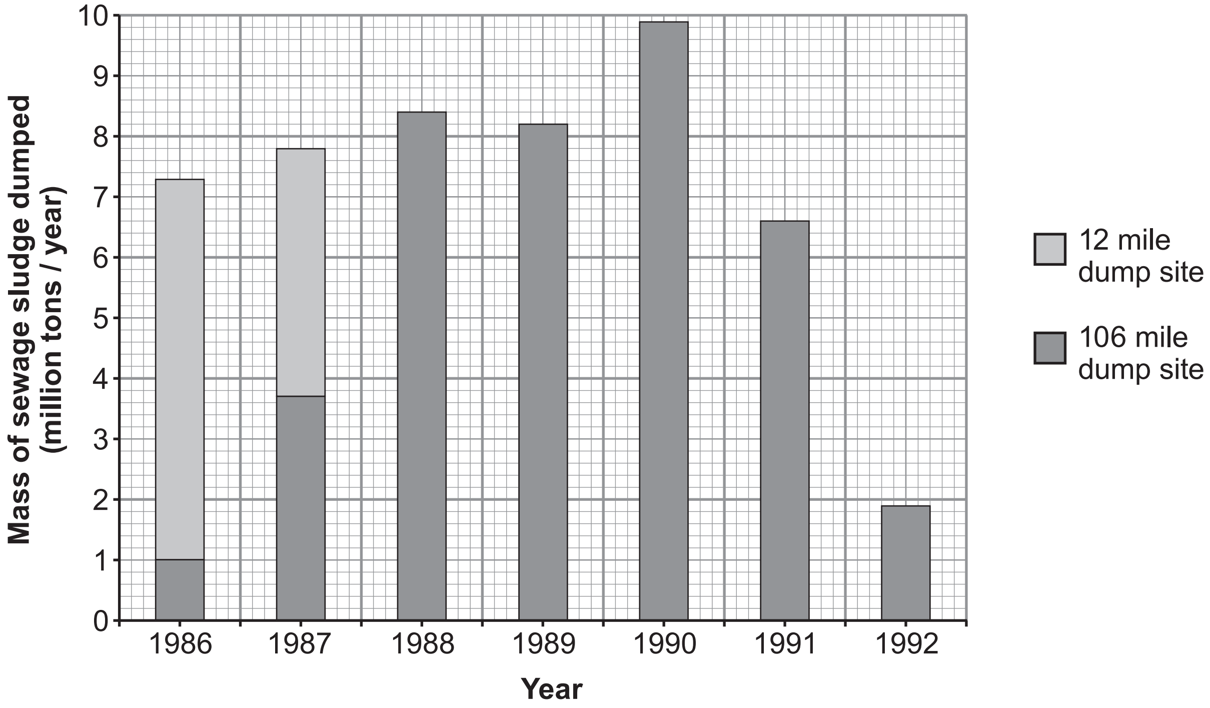
(All data from the US Environmental Protection Agency Report to Congress Sept 1995)

Map **1** showing the east coast of the USA together with disposal sites



Examiner only

Graph showing the annual disposal of sewage sludge at the ‘12 mile’ and ‘106 mile’ dump sites from 1986 to 1992.



(a) (i) Calculate the total mass of sewage sludge disposed of at the ‘12 mile’ dump site in 1986 and 1987. [2]

total mass of sewage sludge = ... million tons

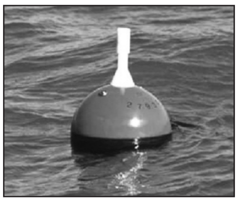
(ii) What was the final year in which sewage sludge was disposed of at the ‘12 mile’ dump site? [1]

....

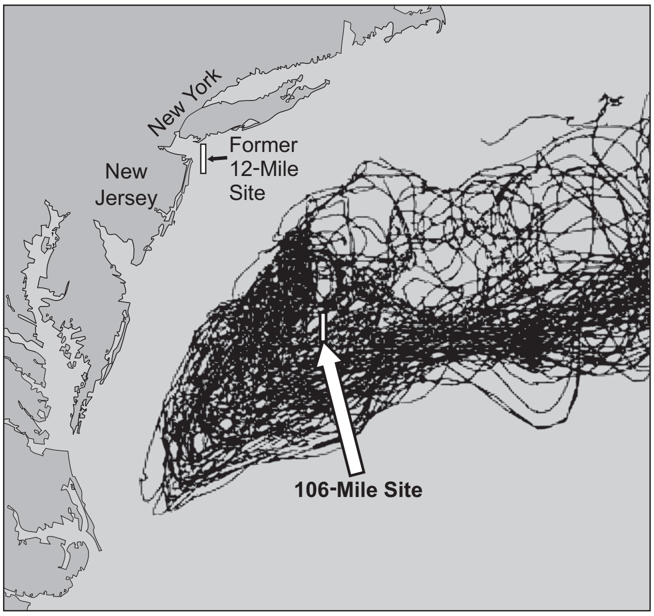
Examiner only

(iii) The US Environmental Protection Agency released buoys into the ocean at the 106 mile dump site. They used satellites to track the movement of the buoys between 1989 and 1992.

drifting buoy tracked by satellite



Map **2** showing the movement of buoys



Use the information in map **2** to suggest why it was decided to select a sewage dump site 106 miles off the east coast of the USA and to close the '12 mile' dump site. [2]

....

....

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Examiner only

(b) Of the sewage sludge which is dumped, 20 – 70% reaches the sea bed. Here the oxygen consumed by living organisms increases greatly. Explain why this happens. [3]

....

....

....

....

....

(c) Close to the '106 mile' sewage dump site is the former '106 mile' **industrial** waste dump site. (See map **1** on page 16). Name **one** group of industrial wastes which you would expect to find at this site. [1]

....

9

END OF PAPER

Surname	Centre Number	Candidate Number
Other Names		0

GCSE

3430U10-1

S19-3430U10-1

FRIDAY, 7 JUNE 2019 – AFTERNOON

SCIENCE (Double Award)

Unit 1: BIOLOGY 1
FOUNDATION TIER

1 hour 15 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**12**	
2.	**11**	
3.	**6**	
4.	**8**	
5.	**8**	
6.	**6**	
7.	**9**	
Total	**60**	

3430U101
01

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

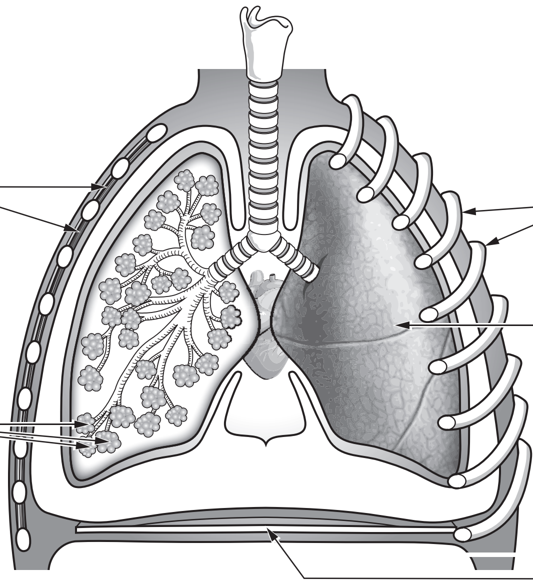
The number of marks is given in brackets at the end of each question or part-question.

Question **4**(*b*)(ii) is a quality of extended response (QER) question where your writing skills will be assessed.

Examiner only

Answer **all** *questions.*

1. *(a)* The diagram shows the human thorax after breathing in.



(i) State the name of the structures labelled:

X .... [1]

Y .... [1]

(ii) The structures labelled **Z** are the sites of gas exchange.
State the name of the structures labelled **Z**. [1]

....

(iii) Complete the following sentences by choosing the correct words from the list. [2]

decreases **contracts** **relaxes** **increases**

When we breathe **out**, the diaphragm .... and becomes dome shaped. As a result, pressure in the thorax .... , so air is forced out of the lungs.

Examiner only

(iv) Choose the letter (**A-C**) that shows the pathway of air from the lungs when we breathe **out**. [1]

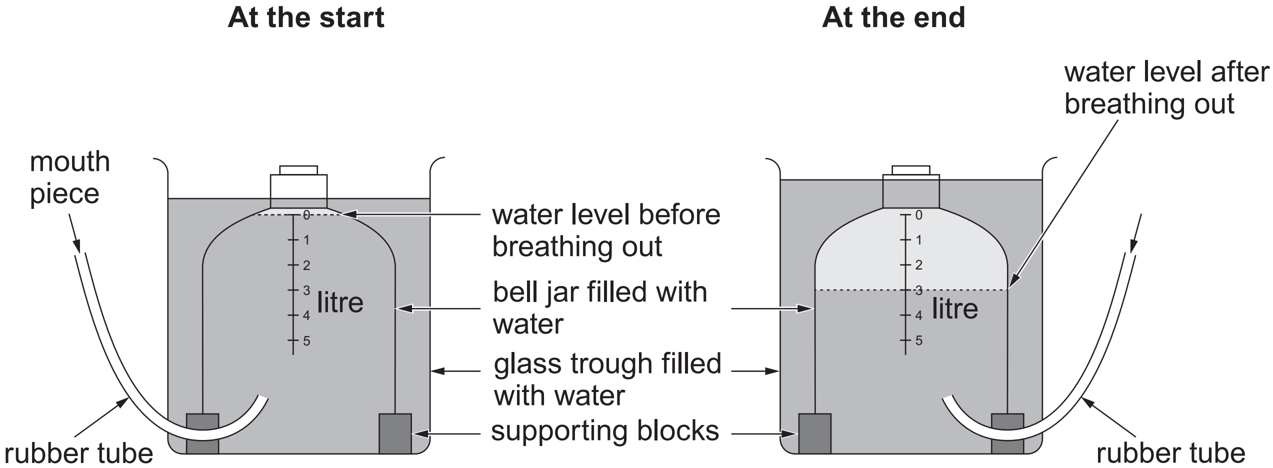
A trachea ⟶ bronchus ⟶ bronchiole

B bronchus ⟶ trachea ⟶ bronchiole

C bronchiole ⟶ bronchus ⟶ trachea

Answer letter = ...

(b) The maximum volume of air that can be breathed out in one breath is called the vital capacity. The diagram shows apparatus that Lowri used to measure her vital capacity.



Describe how Lowri used the apparatus above to measure her vital capacity. [2]

....

....

....

....

....

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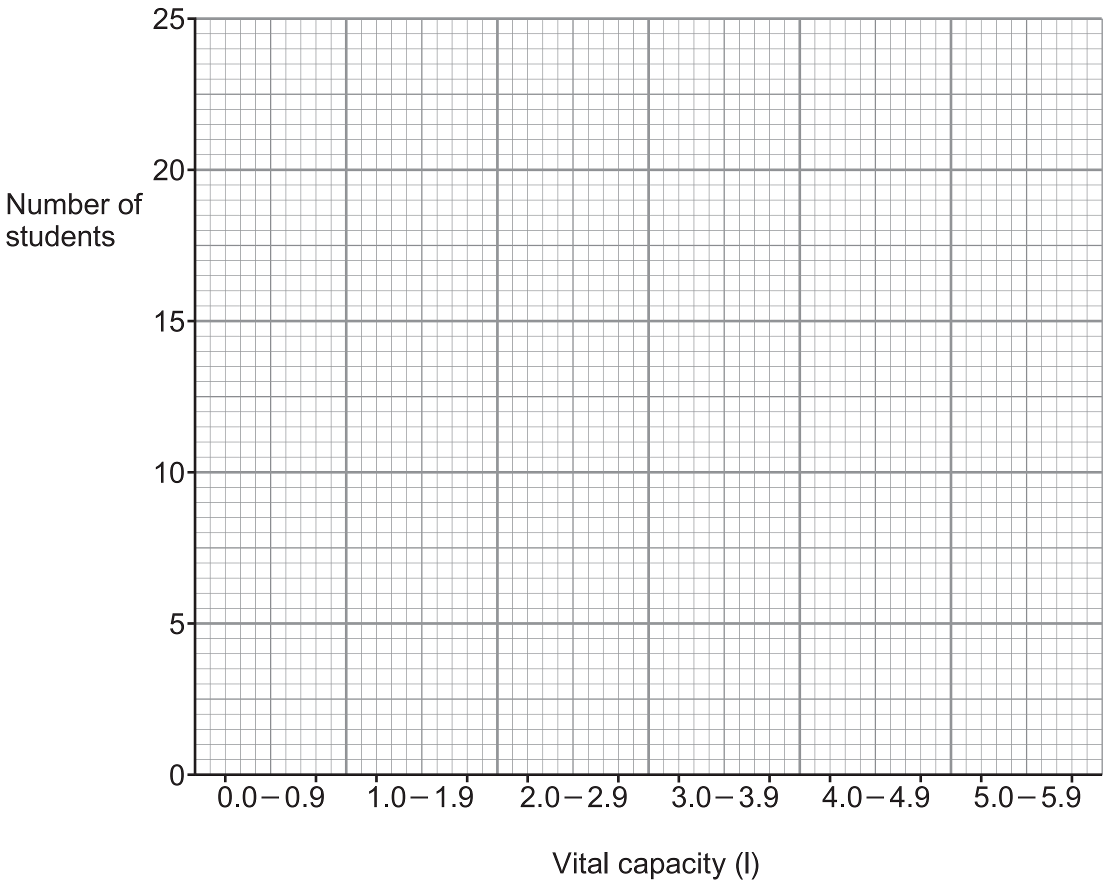
03

Examiner only

(c) The table shows the vital capacities of some students in a school. The vital capacities were measured to the nearest 0.1 litre.

Vital capacity (l)	Number of students
0 - 0.9	5
1.0 - 1.9	9
2.0 - 2.9	20
3.0 - 3.9	8
4.0 - 4.9	23
5.0 - 5.9	7

(i) Use a ruler to draw a bar chart of the results on the grid. [2]



Examiner only

(ii) Suggest **two** possible reasons to account for the variation in vital capacity. [2]

....

....

....

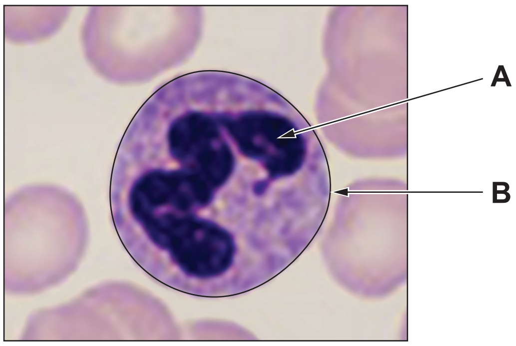
....

12

3430U101 05

Examiner only

2. *(a)* The photograph shows one labelled blood cell from the circulatory system.



(i) Complete the table by stating the name and function of structures **A** and **B**. [4]

Structure	Name	Function
A	....	....
B	....	....

(ii) The cell contains many mitochondria. State the function of mitochondria. [1]

....

(b) Complete the table below by using words **from the list** to identify an **organ**, and a **cell** from the **human circulatory system**. [1]

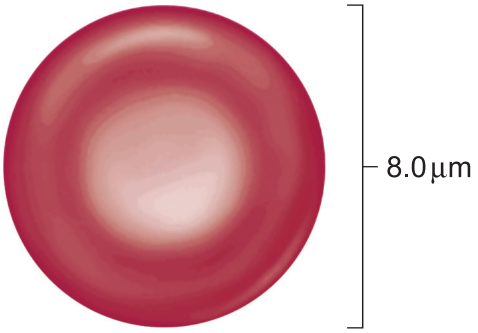
heart **lung** **palisade** **trachea** **phagocyte**

Organ system	Organ	Tissue	Cell
circulatory	....	muscle	....

 (3430U10-1)

Examiner only

(c) The diagram below shows a red blood cell.



(i) I. Complete the following calculation. [1]

diameter of red blood cell = 8.0 µm

radius (r) = 4.0 µm

Radius^2 = ... µm

II. Use your answer to part I. and the formula 3.14 × radius^2, to calculate the surface area of one side of a red blood cell to the **nearest whole number**. [1]

Surface area of one side of red blood cell = ... µm^2

(ii) Red blood cells have a large surface area.
State the function of red blood cells and suggest how a large surface area helps red blood cells to be more efficient. [2]

....

....

....

(iii) Red blood cells are specialised cells.
Give the meaning of the term *specialised cell.* [1]

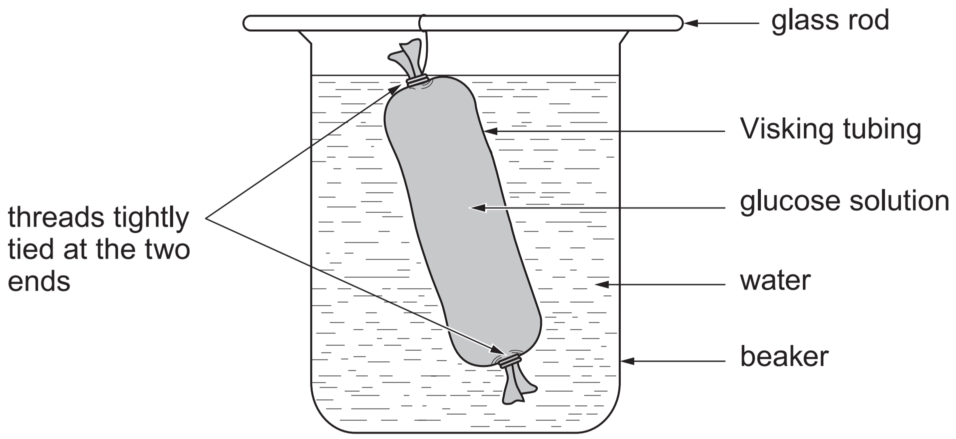
....

3430U101 07

11

Examiner only

3. Visking tubing has small pores (holes) in its membrane. Students investigated the movement of molecules through membranes using Visking tubing.
The students set up the apparatus shown below.



They took samples of water from the beaker at the start and at 15 minutes.

They tested the samples for the presence of glucose using Benedict's reagent.
The results are shown in the table.

Test	Observations	
	at the start	at 15 minutes
Benedict's reagent	blue	brick red

(a) From the table:

(i) State the conclusion that can be made from the Benedict's test observation **at the start**. [1]

....

(ii) Explain the observations at 15 minutes for the Benedict's test. [2]

....

....

....

....

Examiner only

(b) During the investigation, a number of processes occurred.
Use the information opposite to complete the following table by writing **true** or **false** against **each** statement. One has been done for you. [3]

Statement	**True** or **false**
The water in the beaker became a solution.	....
The concentration of the glucose solution in the tubing increased.	....
Osmosis occurred.	true
Water molecules passed through the membrane.	....
The number of water molecules in the tubing increased.	....

3430U101 09

6

Turn over.

Examiner only

4. Lichens can be used as indicators of the level of air pollution.

There are three types of lichen – bushy, leafy and crusty.
The photograph shows the three types of lichen.

bushy leafy crusty

The table below shows how the presence (✓) or absence (x) of each type of lichen in an area allows an assessment to be made of the level of air pollution.

Type of lichen	No pollution	Low pollution	Moderate pollution	High pollution
bushy	✓	x	x	x
leafy	✓	✓	x	x
crusty	✓	✓	✓	x

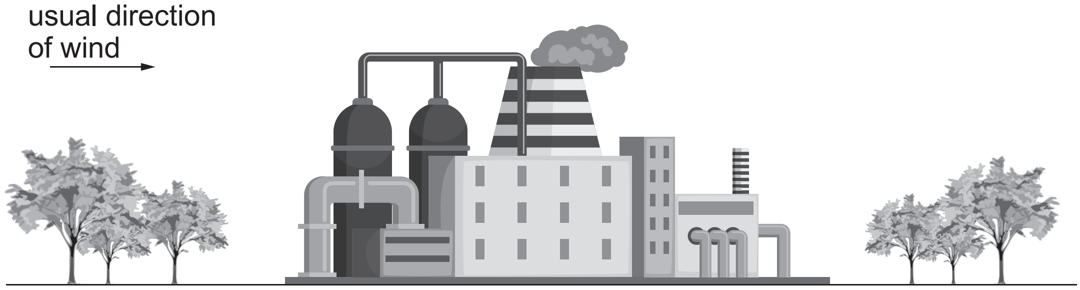
(a) Sharon examined several trees in her school grounds. She used the photograph above to identify each type of lichen she found. She found only leafy and crusty lichens.

State the level of air pollution indicated by Sharon's findings. [1]

....

Examiner only

(b) The drawing shows two woods, **A** and **B**, separated by a factory.



Sharon investigated air pollution in the woods. She made the following hypothesis:
*'Air pollution is higher in wood **B** than in wood **A**'.*

(i) State the evidence in the drawing that Sharon used to make her hypothesis. [1]

....

(ii) Using lichens on trees as indicators, design an investigation to test the hypothesis: *'Air pollution is higher in wood **B** than in wood **A**'.*
Explain how you would use the results to support **or** disprove the hypothesis. [6 QER]

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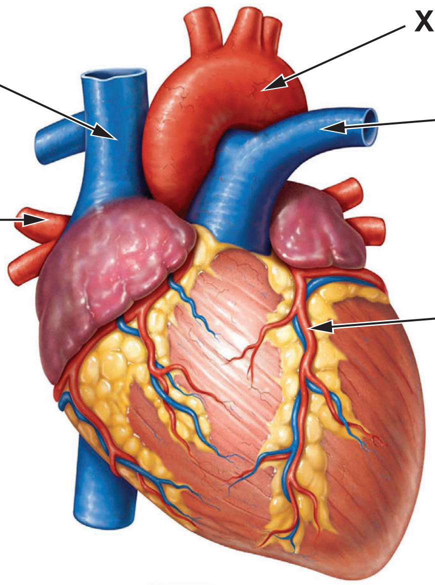
8

Examiner only

5. *(a)* The diagram shows the human heart.



(i) Blood vessel **X** is an artery carrying oxygenated blood from the heart at high pressure.
State the name of blood vessel **X**. [1]

....

(ii) State the function of the coronary artery. [1]

....

(b) Don investigated the effect of increasing intensity of exercise on the heart rate of three boys in his class, Dylan, Dewi and Dougie. His method is shown below.

1. He asked them to walk.
2. He counted their heart rate for one minute.
3. He repeated the method for jogging and then sprinting.

The results are shown in the table.

	Heart rate (beats/minute)		
	walking	jogging	sprinting
Dylan	74	84	96
Dewi	82	88	98
Dougie	80	89	98
Mean	79	87	97

Examiner only

(i) From the table:

I. State the relationship between increasing intensity of exercise and heart rate. [1]

....

II. Explain the importance of the relationship for respiring muscle tissue. [2]

....

....

....

....

....

(ii) Suggest **two** ways Don could improve his experimental method in order to make a **fair** comparison between the three boys. [2]

....

....

....

....

(iii) Suggest **one** way that Don could develop his investigation to make his results more representative of the whole population of Wales. [1]

....

....

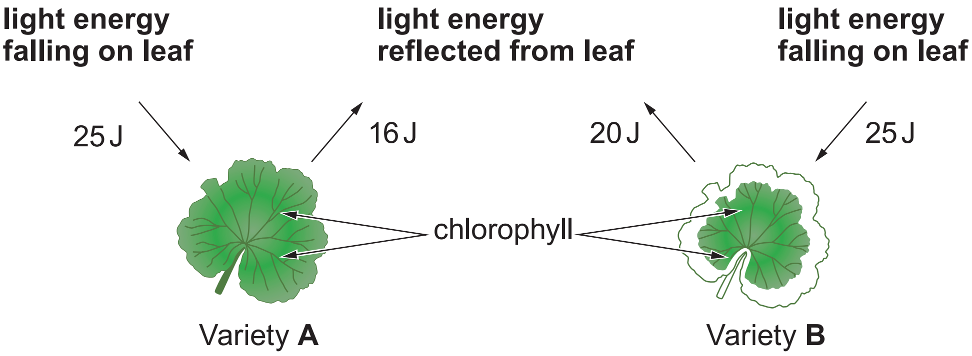
8

Examiner only

6. *(a)* Write the word equation for photosynthesis. [2]

....

(b) The diagram shows leaves from two varieties of geranium **A** and **B** of the genus *Pelargonium*. Light energy falling on the two leaves and the light energy reflected from each leaf, during a period of one hour is also shown.



% light energy reflected

Variety **A** = 64%

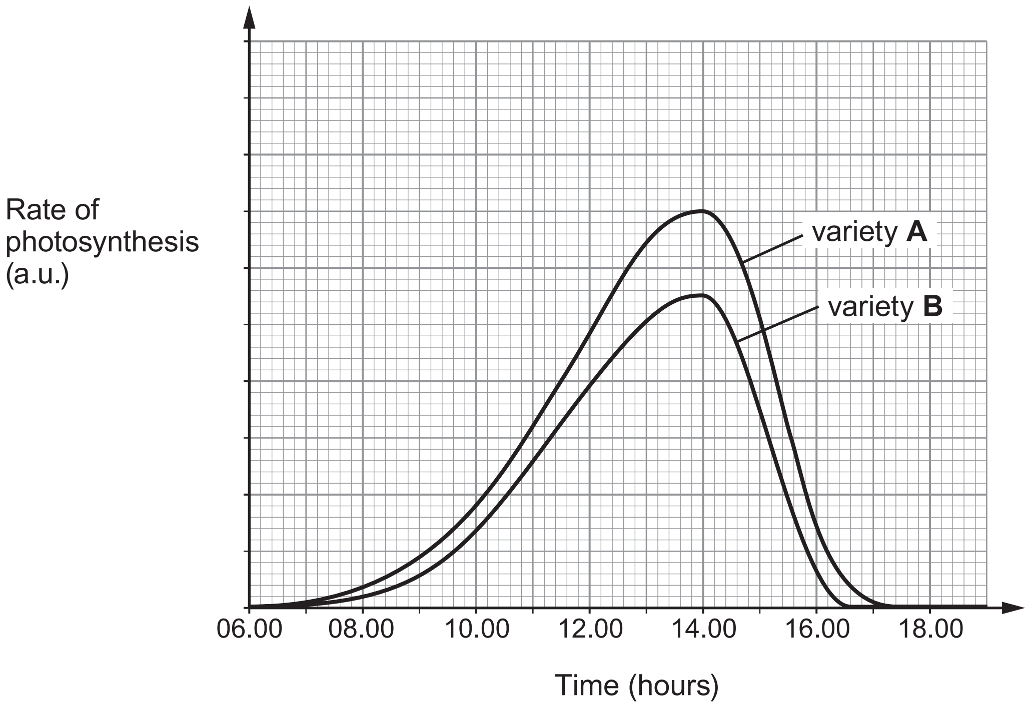
Variety **B** = ... %

Calculate the percentage of light energy falling on leaf **B** that is reflected.
Write your answer in the diagram. [2]

Space for working.

Examiner only

(c) The graph shows the rate of photosynthesis in the two plants between 06.00 hours and 18.00 hours.



Use the diagram opposite and the graph above to explain why the mass of variety **A** is likely to increase at a faster rate than the mass of variety **B**. [2]

....

....

....

....

....

6

Turn over.

Examiner only

7. *(a)* (i) State the products of protein digestion. [1]

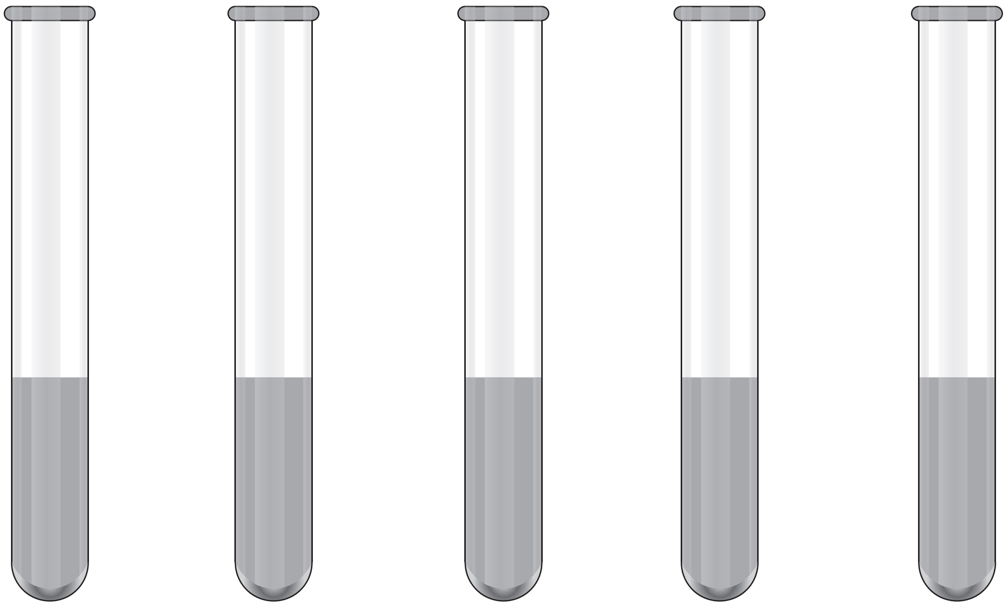
....

(ii) State the part of the digestive system that absorbs digested food molecules. [1]

....

(b) Protein digestion starts in the stomach.

Students investigated protein digestion. They set up five tubes **A** to **E**, as shown below.



Tube **A** **B** **C** **D** **E**

<table>
<tr><th>Tube A</th><th>Tube B</th><th>Tube C</th><th>Tube D</th><th>Tube E</th></tr>
<tr><td>5 cm^3 of 1% protein</td><td>5 cm^3 of 1% protein</td><td>5 cm^3 of 1% protein</td><td>5 cm^3 of 1% protein</td><td>5 cm^3 of 1% protein</td></tr>
<tr><td>5 cm^3 of 0.1% protease</td><td rowspan="3">15 cm^3 liquid from the stomach</td><td>5 cm^3 of 0.1% protease</td><td rowspan="2">15 cm^3 distilled water</td><td rowspan="2">15 cm^3 distilled water</td></tr>
<tr><td>10 cm^3 distilled water</td><td>10 cm^3 distilled water</td></tr>
<tr><td>pH 2.0</td><td>pH 7.0</td><td>pH 2.0</td><td>pH 7.0</td></tr>
</table>

Examiner only

The students measured the percentage protein digested at one hour. The results are shown in the table.

Tube	Percentage protein digested
A	100
B	98
C	5
D	0
E	0

(i) Compare the results for tubes **A** and **D**. State the conclusion that can be made about the digestion of protein. [1]

....

....

(ii) State the conclusions that can be made about the contents of the liquid from the stomach in Tube **B**. [2]

....

....

....

....

(iii) All the 1% protein added to Tube **A** had been fully digested after one hour.
State why the contents of Tube **A** would still test positive for protein. [1]

....

....

(iv) The students carried out a similar investigation, but instead of using 0.1% protease, they used 5 cm^3 of 0.1% lipase in both tubes **A** and **C**.
They found there had been no digestion of protein in either tube.
Use your knowledge of enzyme structure and function to explain the results that the students obtained. [2]

....

....

....

....

(v) State **one** further variable that should be controlled during this investigation. [1]

....

9

END OF PAPER

Surname	Centre Number	Candidate Number
First name(s)		0

GCSE

3430U10-1

Z22-3430U10-1

WEDNESDAY, 15 JUNE 2022 – MORNING

SCIENCE (Double Award)

Unit 1: BIOLOGY 1
FOUNDATION TIER

1 hour 15 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**5**	
2.	**6**	
3.	**10**	
4.	**13**	
5.	**11**	
6.	**9**	
7.	**6**	
Total	**60**	

3430U101
01

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **5**(b) is a quality of extended response (QER) question where your writing skills will be assessed.

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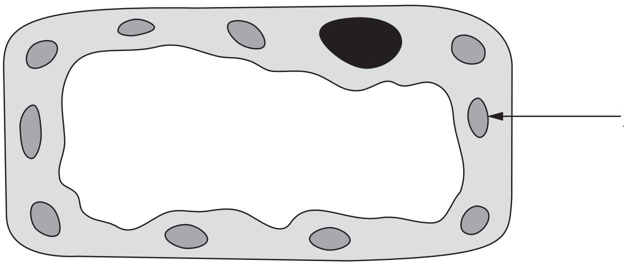
Examiner only

Answer **all** questions.

1. (a) An **incomplete** diagram of a plant leaf cell is shown in **Image 1.1**.

Image 1.1



(i) **Complete Image 1.1 by drawing** the **cell wall**. [1]

(ii) Structure **X** contains chlorophyll and absorbs light.

I. **Complete the following sentence** by underlining the correct term in the brackets. [1]

Structure **X** is a **(vacuole / chloroplast / mitochondrion)**.

II. State the name of the **process** that uses the light absorbed by structure **X**. [1]

……………………………………………………………………

(b) The cell in **Image 1.1** is adapted for one function.

Complete the following sentences by underlining the correct term in the brackets. [2]

(i) Cells adapted for one function are called **(special / specialised / specific)**.

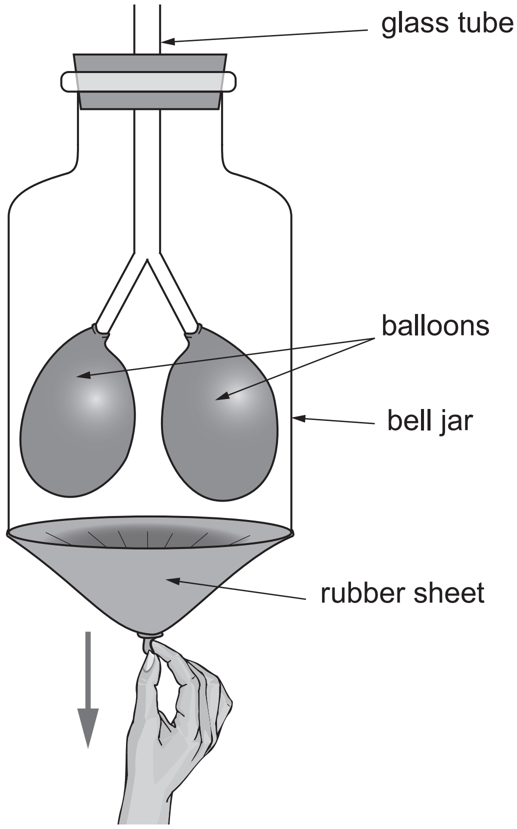
(ii) Groups of similar cells are called **(tissues / organs / organisms)**.

5

Examiner only

2. The bell jar model in **Image 2.1** is used to show how breathing occurs.

Image 2.1



(a) **Use words from the list** to complete the following sentences. [3]

inflates **decreases** **deflates** **equalises** **increases**

When the rubber sheet is pulled down the volume of the space inside the bell jar

………………………………………… and the air pressure ………………………………………… .

As a result, air is forced down the glass tubes and each balloon ………………………………………… .

(b) The equipment labelled on the bell jar model in **Image 2.1** represent different structures in the body. Name the structures represented by:

(i) the glass tube; ………………………………………………………… [1]

(ii) the bell jar; ………………………………………………………… [1]

(iii) the rubber sheet. ………………………………………………………… [1]

6

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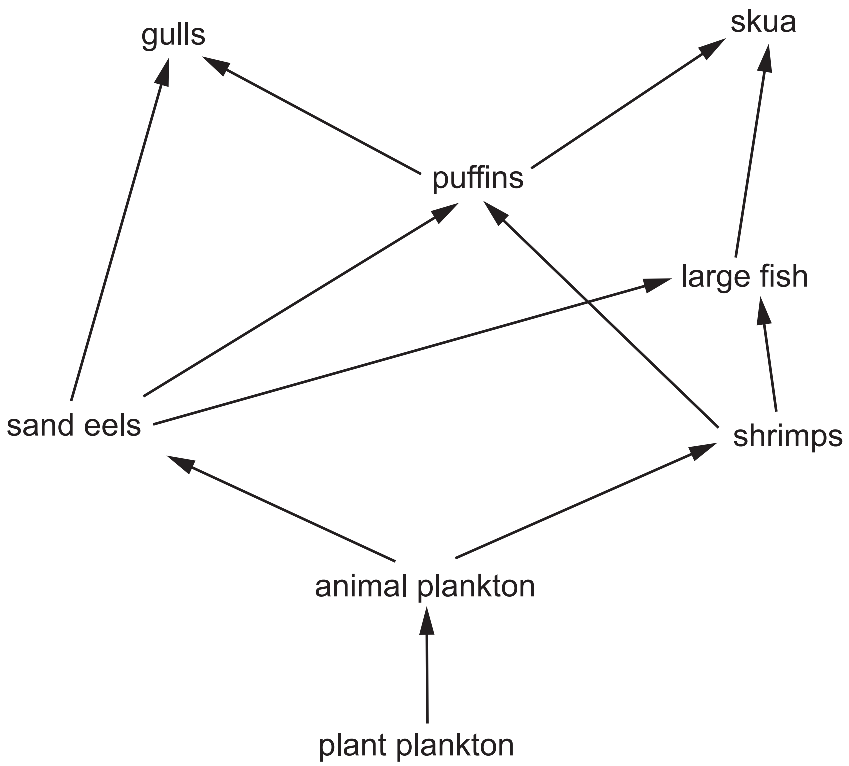
03

Examiner only

3. **Image 3.1** shows part of a marine food web.

Image 3.1



(a) Use the food web in **Image 3.1** to complete a food chain below which **includes puffins**. [3]

plant plankton → .... → .... → puffins → ....

(b) (i) State the **source** of the energy that enters the plant plankton. [1]

....

(ii) Energy is used by the organisms in the food web.
State **one** way energy is used by organisms. [1]

....

Examiner only

(c) Read the following information about puffins.

- Puffins are seabirds.
- Each year, puffins spend eight months at sea. The other four months are spent on land during the breeding season.
- Puffins nest on the ground where there may be many predators such as foxes and rats.
- Natural factors cause puffin numbers to vary. However, oil pollution and rising sea temperatures from climate change have reduced puffin numbers in most areas.
- Some puffins breed on the Welsh island of Skomer. Here, their numbers increased from 14 000 in 2013 to 31 000 in 2018.

Skomer Island

walesonline.co.uk

Use the above information to answer the following questions.

(i) Explain why the Welsh Wildlife Trusts prevent rats from being introduced onto Skomer. [1]

....

(ii) **Complete the table** by writing true or false for each statement about puffins. [4]

Statement	True or False
Puffins face predators only at sea.	False
Puffin numbers are affected by variations in natural factors.	....
Puffin numbers generally are rising.	....
Puffin numbers on Skomer increased by over 100% between 2013 and 2018.	....
Puffins are at risk from climate change.	....
Puffins spend only one third of the year at sea.	....

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10

Examiner only

4. (a) The photograph shows a fitness smartwatch.

Bethan used a smartwatch to investigate the resting heart rates of ten students in her class.

- Five girls and five boys were chosen at random.
- Each student sat at rest for one minute.
- Then the heart rate shown on the smartwatch was recorded.

The results are shown in **Table 4.1**.

Table 4.1

Girls		Boys	
Name	Resting heart rate (beats per minute)	Name	Resting heart rate (beats per minute)
Seren	69	Dan	62
Katya	74	Jim	65
Nia	73	Ifor	67
Tracy	59	Rhys	60
Angharad	70	Mohamed	63
	mean = 69		mean = ...

Examiner only

(i) I. Calculate the mean resting heart rate for the boys.
Give your answer to the nearest whole number. Write your answer in Table 4.1. [3]
Space for working.

II. State the conclusion that can be made from a comparison of the two means. [1]

....

....

(ii) State **one** way that this investigation is a fair test. [1]

....

....

3430U101
07

(b) **Table 4.2** shows the mean resting heart rates by age in women and men from thousands of fitness smartwatch users of many nations and ethnicities.

Table 4.2

Age (years)	Mean resting heart rate (beats per min)	
	Women	Men
20	67.0	62.5
30	67.5	63.5
40	68.0	64.0
50	67.0	64.5
60	66.0	64.0
70	65.0	62.0
80	64.0	61.0

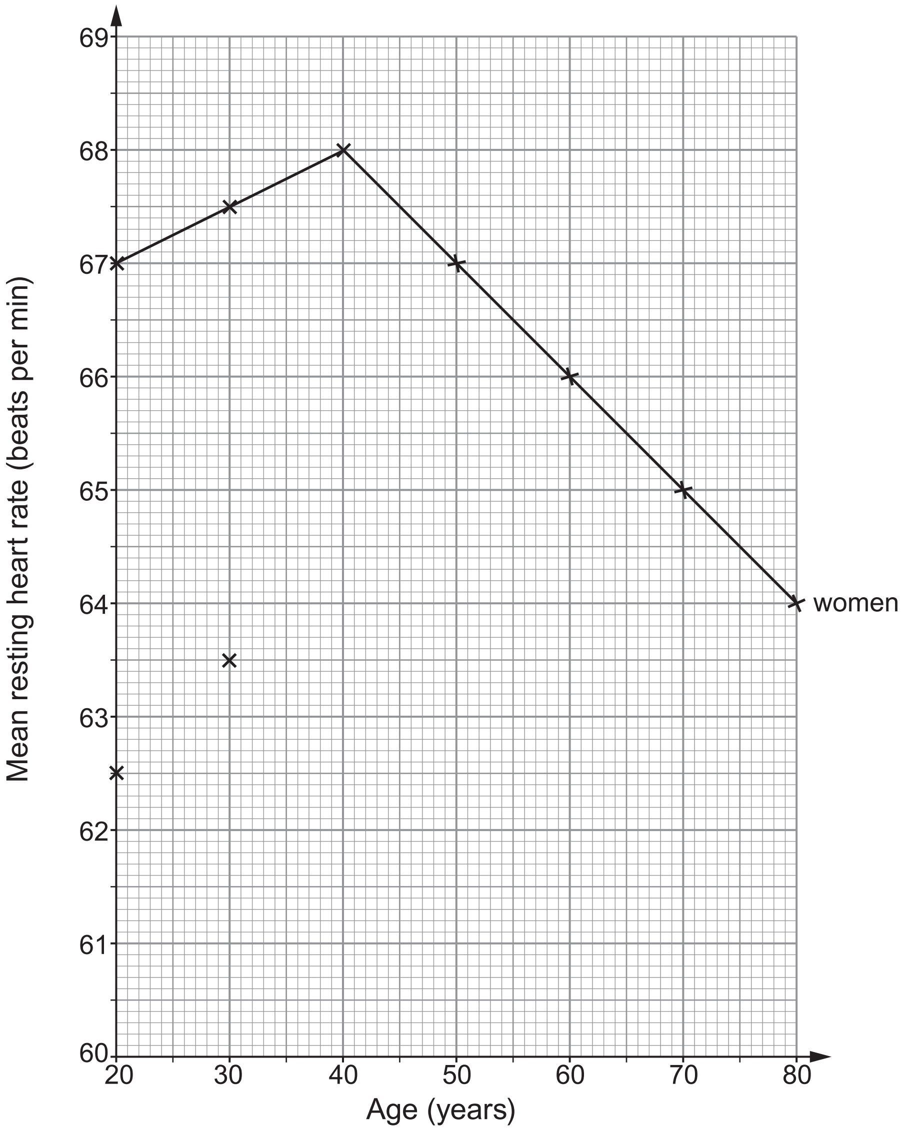
(i) **Graph 4.3** shows the plotted data for women and two plots for men. Complete **Graph 4.3** by: [3]

I. plotting the remaining points for the **men** on the grid.

II. joining **all** the plots **for men** with a ruler.

Examiner only

Graph 4.3



(ii) From **Graph 4.3**:

State whether the data support the results of Bethan's investigation and explain your answer. [1]

....

....

(iii) Describe how the mean resting heart rate in **men** changes between the ages of 20 and 80 years. [2]

....

....

....

....

Examiner only

(c) Give **two** reasons why the smartwatch data in **Table 4.2** are more representative of mean human resting heart rates than those in **Table 4.1**. [2]

1. ....

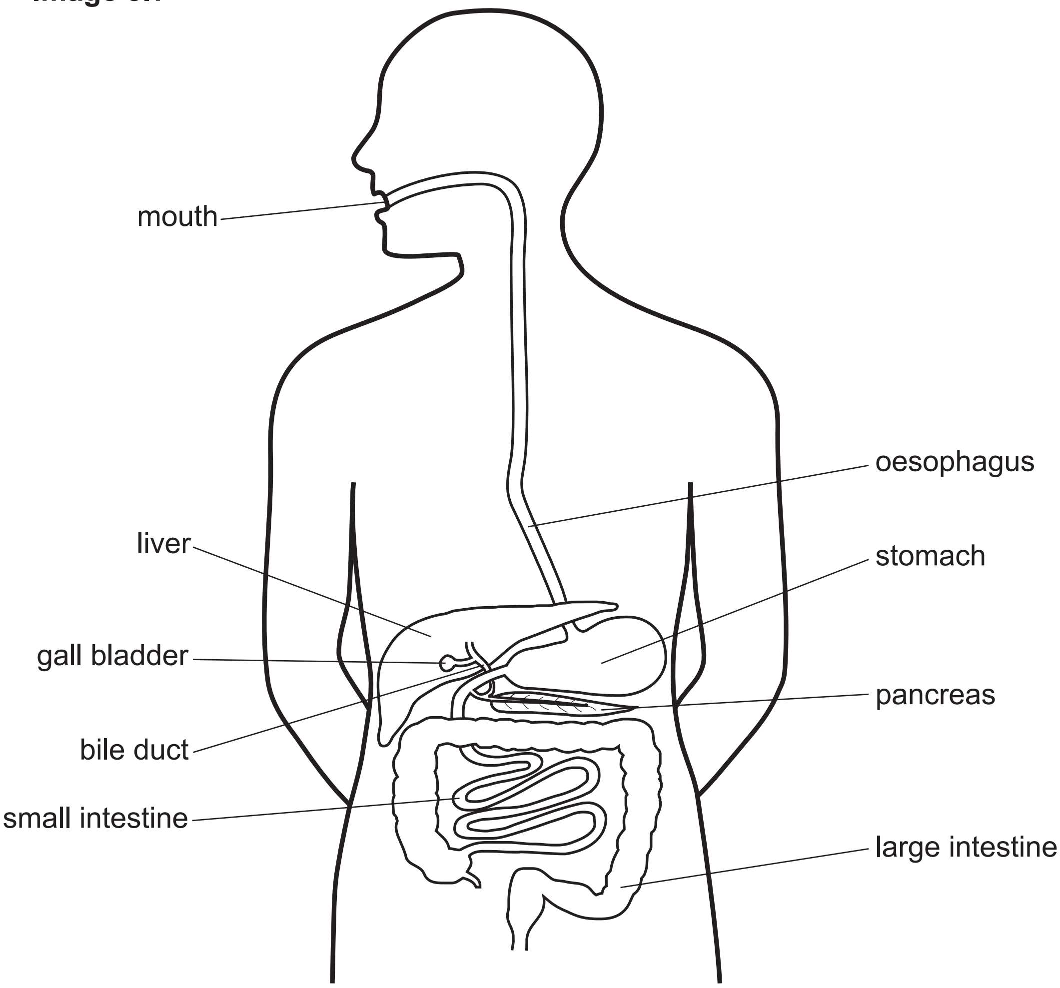
2. ....

13

Examiner only

5. (a) **Image 5.1** shows the digestive system.

Image 5.1



Choose named structures from **Image 5.1** to complete **Table 5.2**. [5]

Table 5.2

Function	Name of structure
Starts digestion of starch	....
Carries bile from gall bladder	....
Absorbs water from undigested food waste	....
Absorbs digested food molecules into the blood	....
Makes lipase	....

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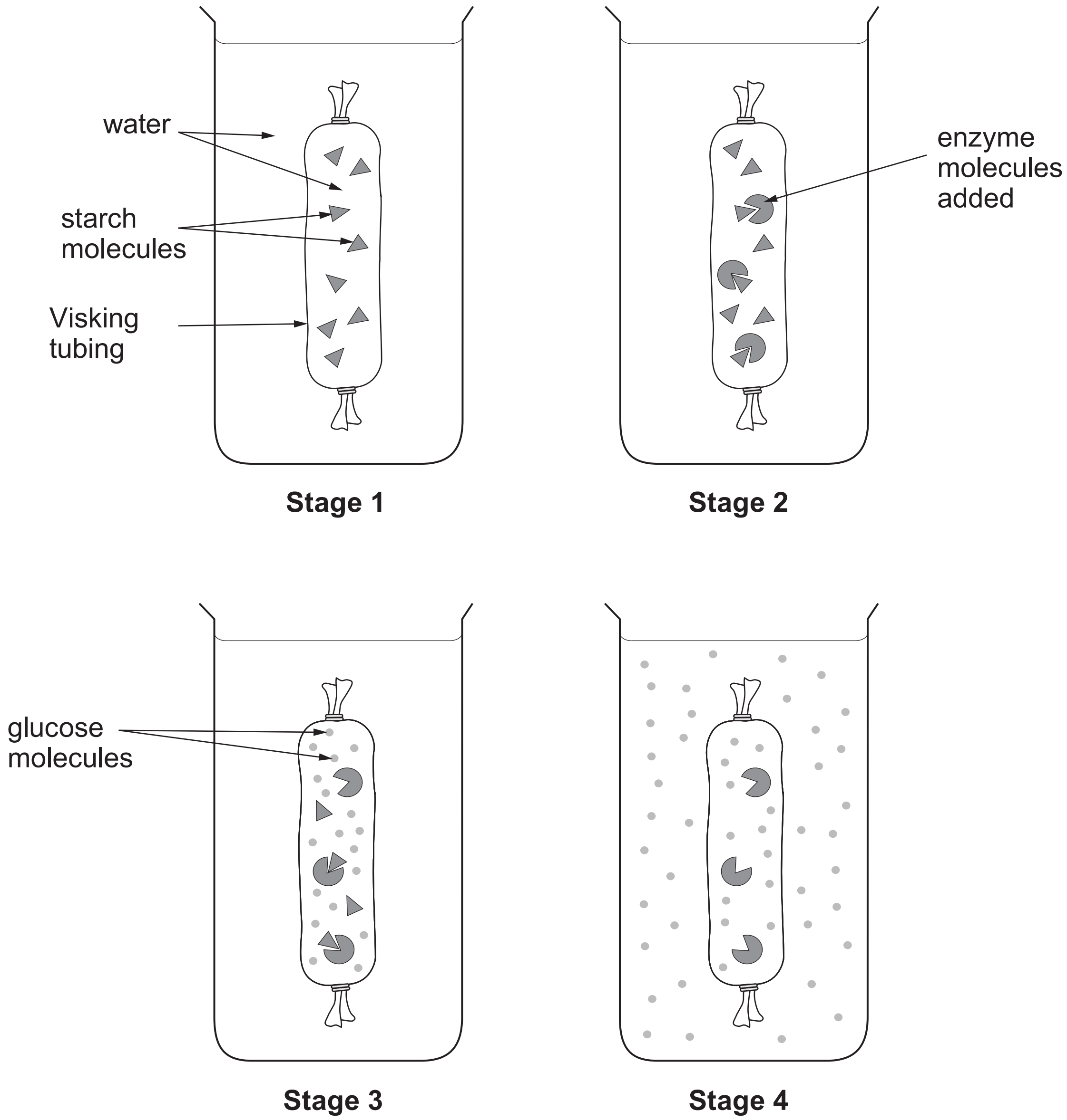
(b) Visking tubing has small holes in its wall which only allow small molecules to pass through.

Dafydd studies a computer model which uses Visking tubing to show part of the digestive system in action.
The stages involved are shown in **Image 5.3**.

Image 5.3



Using **all** the information shown in **stages 1, 2, 3 and 4**, explain how glucose appears in the water outside the Visking tubing in **stage 4**. [6 QER]

....

....

....

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6. The nutrition information in **Table 6.1** is taken from a pack of dried pasta.

Table 6.1

TYPICAL VALUES	Per 100 g (of dried pasta)
Energy	761 kJ
Fat	1.3 g
of which – saturated	0.1 g
of which – unsaturated	.... g
Carbohydrates	25.0 g
of which – sugars	2.3 g
Fibre	7.7 g
Protein	13.0 g
Salt	0.05 g

Examiner only

(a) (i) Calculate the value for unsaturated fats. **Write your answer in Table 6.1**. [1]
Space for working.

(ii) State the name of the nutrient which makes up most of the carbohydrates in the dried pasta. [1]

....

(iii) State the importance of a low-salt diet. [1]

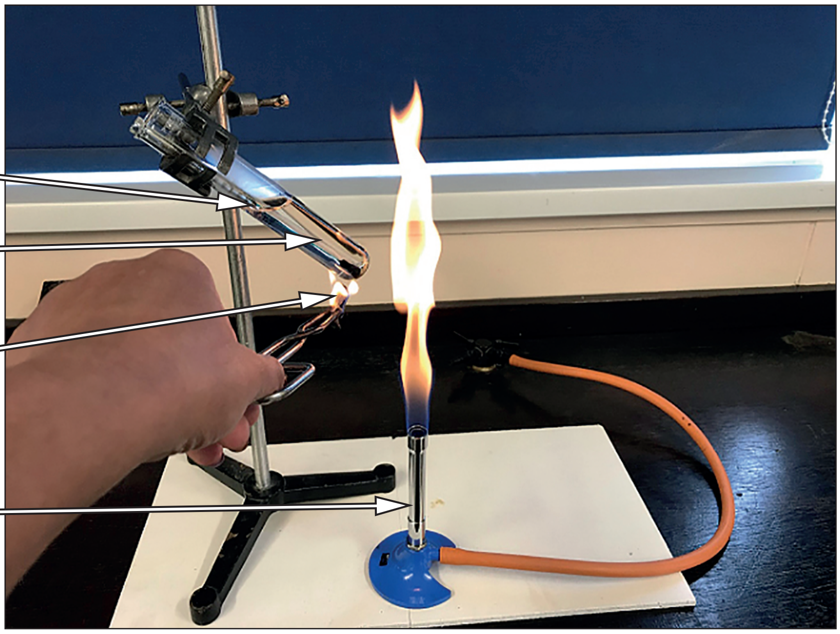
....

....

Examiner only

(b) Lloyd and Emma carried out an experiment to compare the energy values in **Table 6.1** with values they obtained using the apparatus shown in **Image 6.2**.

Image 6.2



They ignited a 1.6 g piece of dried pasta using the Bunsen burner and immediately held the burning pasta at the base of the boiling tube until it stopped burning. The results Lloyd and Emma obtained are shown in **Table 6.3**.

Table 6.3

Mass of pasta (g)	Initial temperature of water (°C)	Final temperature of water (°C)	Increase in temperature of water (°C)	Energy released per gram of food (kJ)
1.6	14	58	44	....

(i) Use the following formula to calculate the energy released per gram of food (kJ). **Write your answer in Table 6.3**. [2]

$$\text{Energy released per gram (kJ)} = \frac{\text{volume of water (cm}^3\text{)} \times \text{temperature increase (°C)} \times 0.0042}{\text{mass of pasta sample (g)}}$$

Space for working.

Examiner only

(ii) I. State how the energy content of dried pasta in **Table 6.3** compares to the energy content indicated in **Table 6.1**. You must use numerical data in your answer. [2]

....

....

....

....

II. Give **one** reason for the difference between the energy content of dried pasta obtained by Lloyd and Emma, as shown in **Table 6.3** and the energy content indicated in **Table 6.1**. [1]

....

....

(iii) Evaluate the arrangement of the apparatus shown in **Image 6.2** by identifying **one** source of error. [1]

....

....

....

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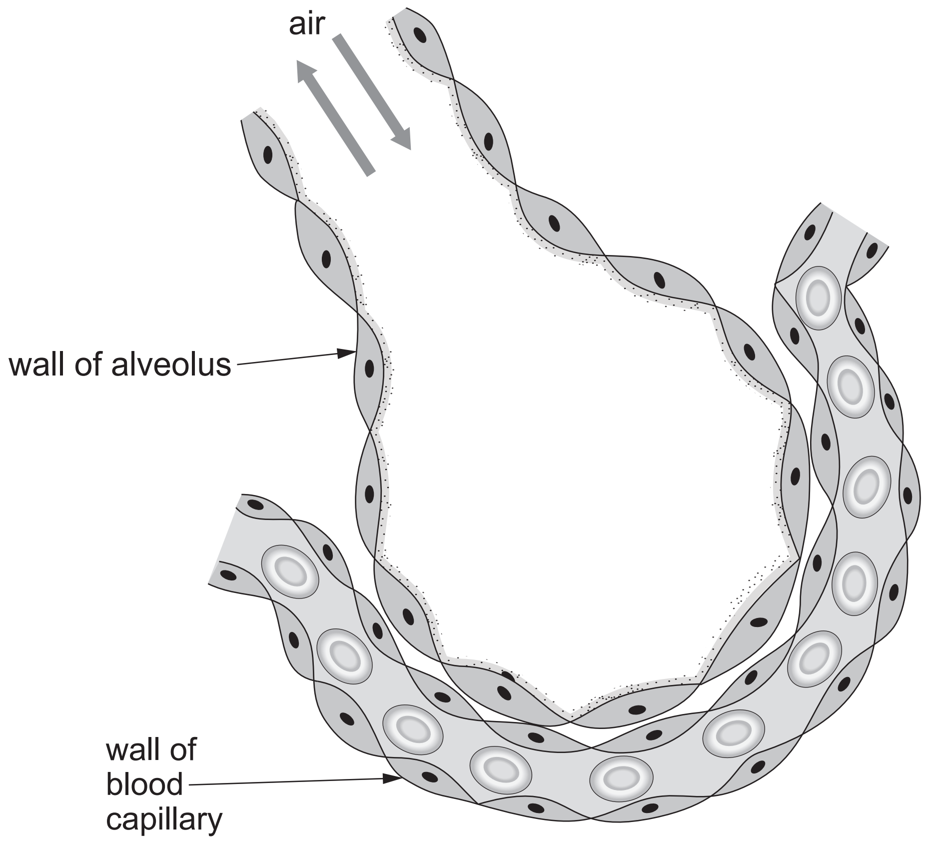
7. **Image 7.1** shows an alveolus.

Image 7.1



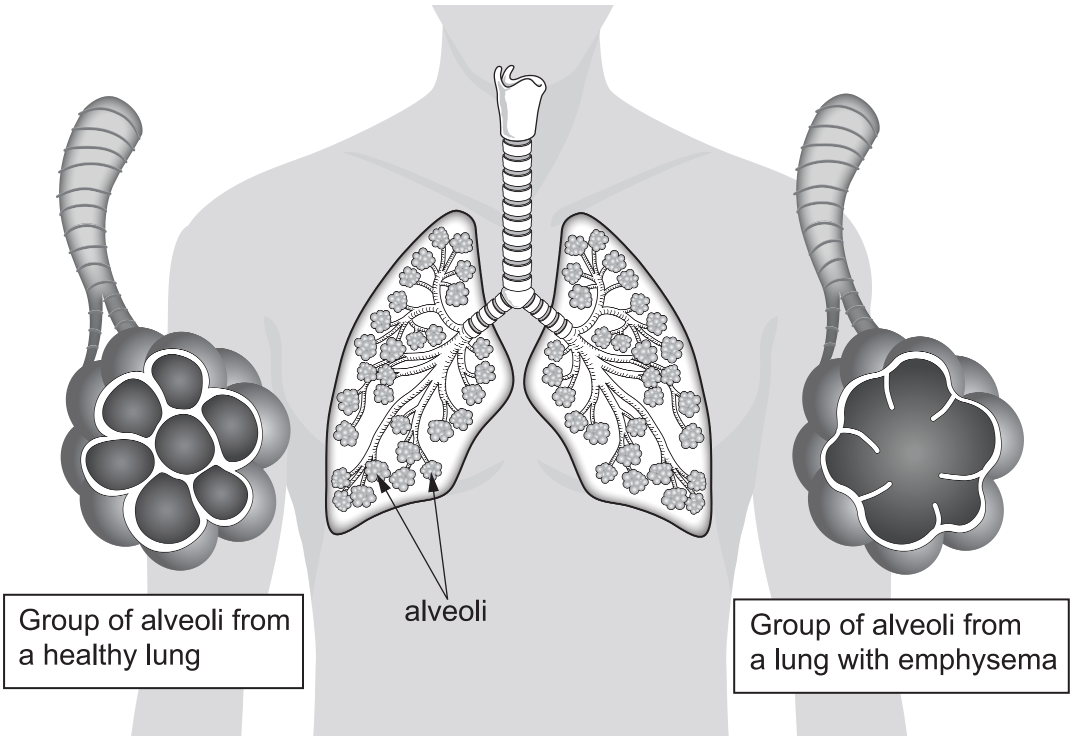
(a) **Use labelled arrows** to identify the following structures on **Image 7.1**: [2]

(i) bronchiole;

(ii) blood plasma.

(b) **Image 7.2** shows groups of alveoli from a healthy lung and from a lung of a person with emphysema. The tables below **Image 7.2** show the concentrations of oxygen and carbon dioxide in the blood capillaries.

Image 7.2



Healthy lung	
Gas	Concentration of gases in the blood capillaries (arbitrary units)
oxygen	86
carbon dioxide	45

Lung with emphysema	
Gas	Concentration of gases in the blood capillaries (arbitrary units)
oxygen	53
carbon dioxide	62

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(i) Using **Image 7.2**, explain the differences in the concentrations of gases in the blood capillaries of a healthy lung and a lung with emphysema. [2]

....

....

....

....

(ii) State the effect on breathing of the difference in concentrations of these gases for a person suffering from emphysema. [1]

....

(iii) State **one** cause of emphysema. [1]

....

6

END OF PAPER

Surname	Centre Number	Candidate Number
First name(s)		0

GCSE

S23-3430U10-1

3430U10-1

TUESDAY, 13 JUNE 2023 – MORNING

SCIENCE (Double Award)
Unit 1: BIOLOGY 1
FOUNDATION TIER

1 hour 15 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**9**	
2.	**11**	
3.	**8**	
4.	**11**	
5.	**6**	
6.	**5**	
7.	**10**	
Total	**60**	

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ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use a pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **5** is a quality of extended response (QER) question where your writing skills will be assessed.

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Answer **all** questions.

1. (a) Plant leaves absorb light for photosynthesis.

State the source of the light absorbed by the leaves. [1]

....

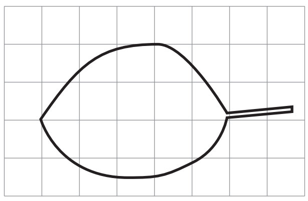
(b) **Image 1.1** shows a violet plant (*Viola riviniana*).

Image 1.1

Gareth calculated the area of a leaf from one of the violet plants. This is what he did:

Step 1. He drew an outline of a leaf on a grid as shown in **Image 1.2**.

Image 1.2



Step 2. He counted the number of **whole** squares inside the outline.

Step 3. He multiplied the total number of whole squares by the area of one square.

Examiner only

(i) Count the number of **whole** squares inside the outline. [1]

Number of **whole** squares inside the outline = ...

(ii) Each square has an area of 0.25 cm^2.
Use the formula below to calculate the outline area. [1]

Outline area = 0.25 × number of whole squares inside the outline.

Outline area = ... cm^2

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(iii) Counting only the whole squares gave an outline area which was much lower than the actual leaf area.

I. Explain why. [1]

....

II. Suggest **one** improvement to **step 2** to get an answer closer to the true value. [1]

....

....

Examiner only

(iv) Gareth decided to test the following hypothesis:

"The leaf area of violet plants in the shade will be greater than that of violet plants in bright light".

He picked one violet plant growing in shade and one growing in bright light.

Gareth used his method to calculate the leaf areas of 10 leaves from each plant.

The mean results are shown in **Table 1.3**.

Table 1.3

Plant growing in	Mean leaf area (cm^2)
shade	1.5
bright light	0.9

I. Calculate the difference in mean leaf area between the two plants. [1]

Difference = ... cm^2

II. **Complete the following sentence** by underlining the correct word. [1]

The results show that Gareth's hypothesis is:

invalid / rejected / supported.

(c) Explain the advantage of a large leaf area for plants growing in shade. [2]

....

....

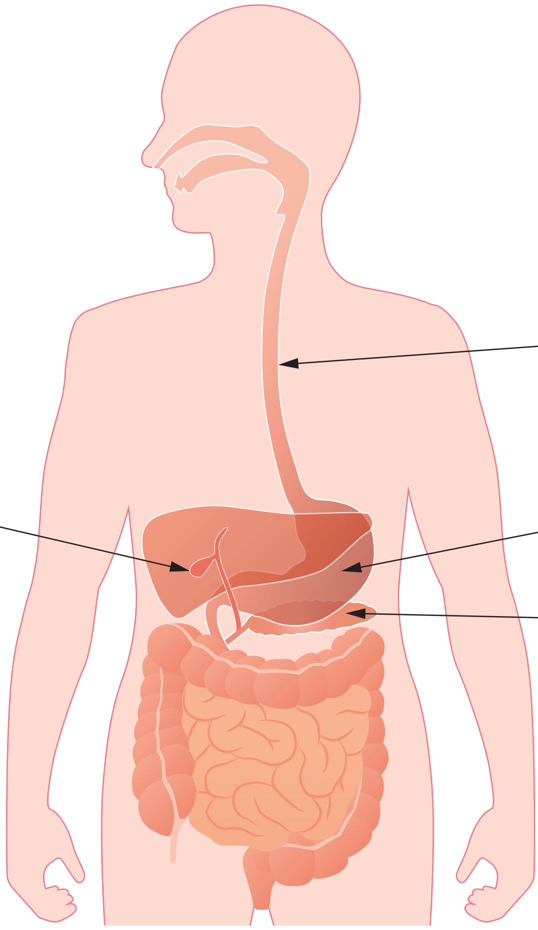
....

9

Examiner only

2. **Image 2.1** shows the digestive system. Some structures have been labelled.

Image 2.1



(a) **Complete the following sentences** by underlining the correct word.

(i) Peristalsis in the digestive system causes food to be

digested / moved / absorbed. [1]

(ii) One organ where peristalsis happens is the

gall bladder / oesophagus / pancreas. [1]

(b) State the name of the organ labelled in **Image 2.1** in which the chemical digestion of protein **starts**. [1]

....

(c) (i) The liver makes bile.
On Image 2.1, draw an arrow to label the liver. [1]

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(ii) Bile helps us digest fat.
Complete Table 2.2 by writing **true** or **false** against **each** statement. One has been done for you. [3]

Table 2.2

How bile works	True or false
increases the number of fat molecules	false
turns large droplets of fat into smaller droplets	....
increases the pH in the small intestine	....
digests fat molecules	....
neutralises stomach acids in the small intestine	....

(d) Fat is an important part of the diet.

State **one** function of fat in the body. [1]

....

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(e) Eating excess fat can lead to obesity.

Chitosan is a product that can help people lose weight.

Chitosan **stops the digestion** of fat.

(i) Explain why taking Chitosan would increase the amount of fat in the waste leaving the body. [2]

....

....

....

....

(ii) Suggest **one** reason why taking Chitosan as part of a diet might be harmful to health. [1]

....

....

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Examiner only

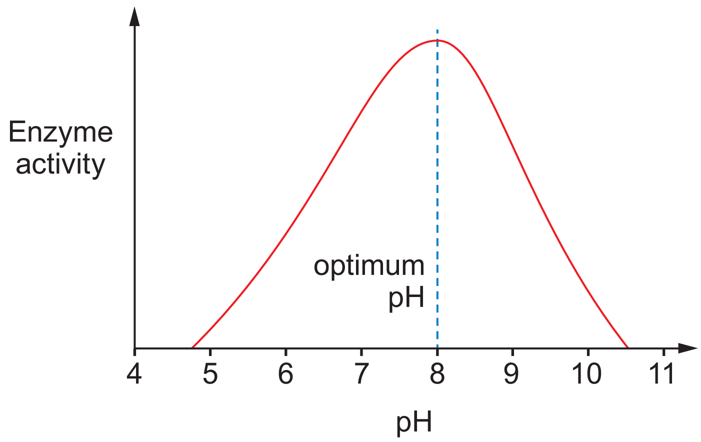
3. This question is about enzymes.

(a) **Complete the following sentence** by underlining the correct word. [1]

Enzymes are made of **carbohydrate / lipid / protein**.

(b) **Graph 3.1** shows the effect of increasing pH on the rate of activity of an enzyme.

Graph 3.1



Use **Graph 3.1** to describe how increasing pH affects enzyme activity. [2]

....

....

....

....

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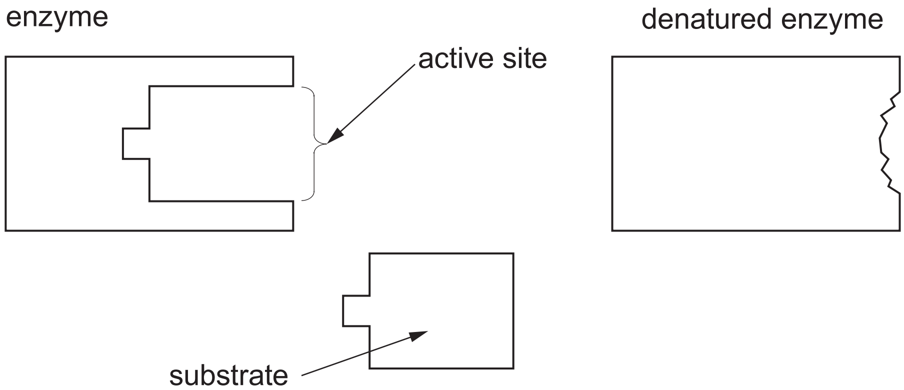
(c) **Image 3.2** shows an enzyme, a denatured enzyme and a substrate molecule.

Image 3.2



(i) Describe what has happened to the active site in the denatured enzyme. [1]

....

....

(ii) Enzymes react with substrates to produce products.
Explain why the denatured enzyme can no longer produce products. [2]

....

....

....

....

(d) State **two** variables that could denature an enzyme. [2]

....

8

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4. (a) **Choose words from the list below to complete the word equation** for aerobic respiration. [2]

carbon dioxide **protein** **oxygen** **nitrogen**

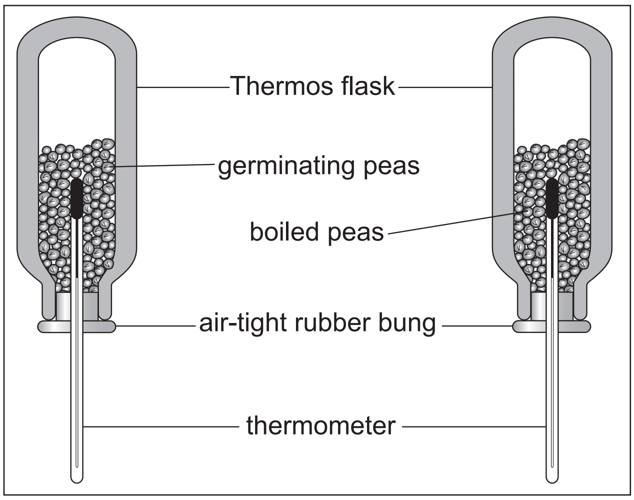
glucose + ... ⟶ .... + water + energy

(b) Tracy investigated the release of energy during respiration in germinating peas using one of the two methods (**1** or **2**) shown in **Image 4.1**.

Image 4.1

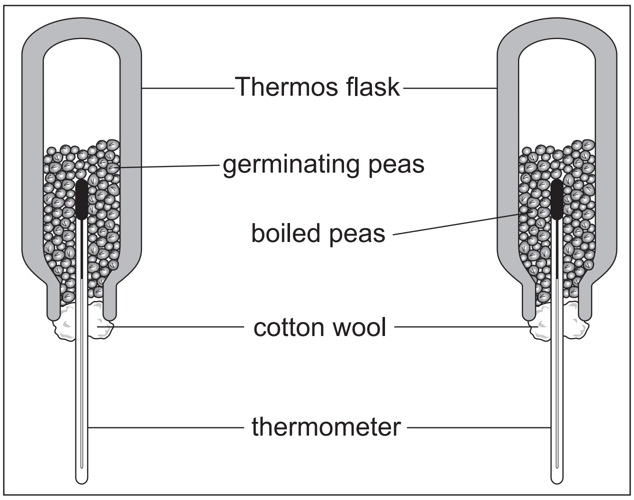
Method 1

Flask **A** Flask **B**



Method 2

Flask **A** Flask **B**



(i) State which of the two methods (**1** or **2**) Tracy should have used and explain your answer. [2]

Tracy should have used Method ....

Explanation

....

....

....

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(ii) **Table 4.2** shows the results of Tracy's investigation over seven days.

Table 4.2

Day	Temperature (°C)	
	flask **A**	flask **B**
1	20.0	20.0
2	21.0	20.0
3	23.0	20.0
4	26.0	20.0
5	27.5	20.0
6	28.5	20.0
7	29.5	20.0

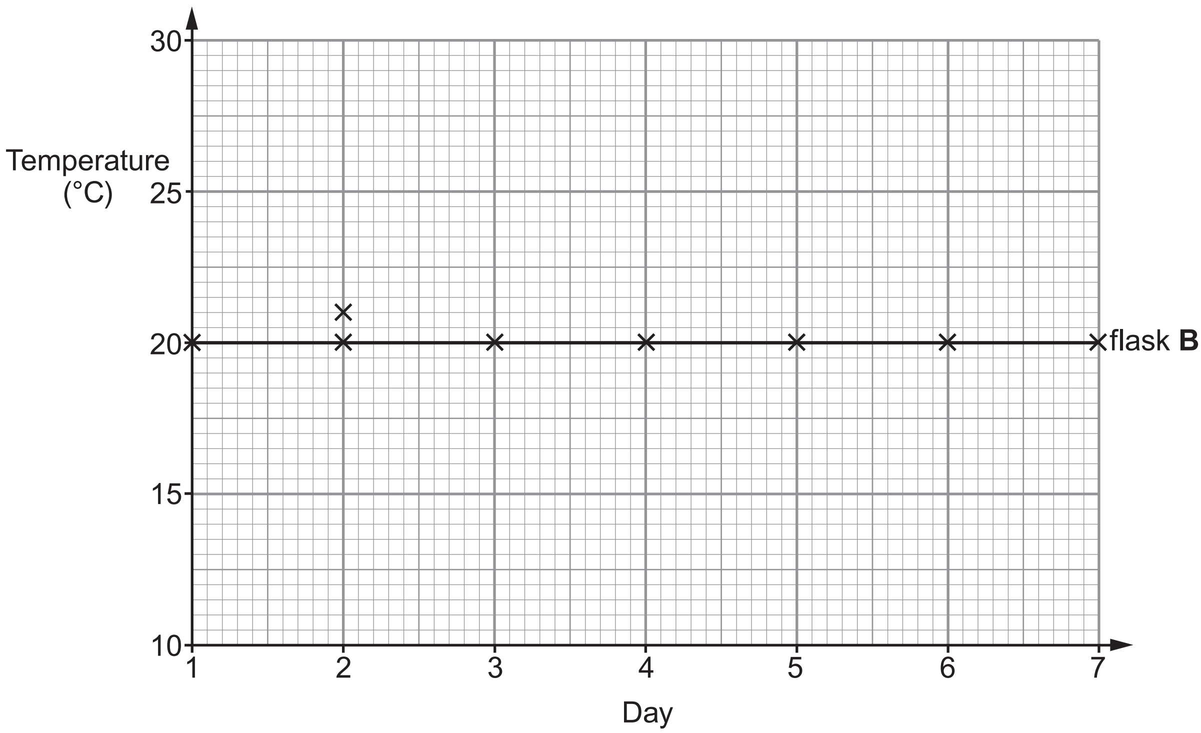
Draw a line graph of the results on **Graph 4.3** by: [3]

I. plotting the points for flask **A** from day 3 to day 7. The first two days have been plotted for you.

II. joining **all the** plots with a ruler and labelling your line as flask **A**.

The results for flask **B** have been plotted for you.

Graph 4.3



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(iii) At the start of the investigation, Tracy made two predictions:

1. The temperature of the peas in flask **A** will rise but the temperature of the peas in flask **B** will not change.

2. Eventually the temperature in flask **A** will stop rising.

Use the results to state whether each of Tracy's predictions were correct. Explain your answers. [2]

Prediction **1** ....

....

Prediction **2** ....

....

(iv) Suggest **one** practical way in which Tracy could confirm or reject her second prediction. [1]

....

....

(v) Tracy recorded the mass of the peas in flask **A** at the start and at the end of the investigation. **Complete the sentence below** by underlining the correct outcome that you would expect Tracy to observe. [1]

During the investigation, the mass of the peas in flask **A** will have:

decreased **increased** **remained the same**

11

Examiner only

5. The beaker in **Image 5** contains 150 cm^3 of an unknown solution.

Image 5

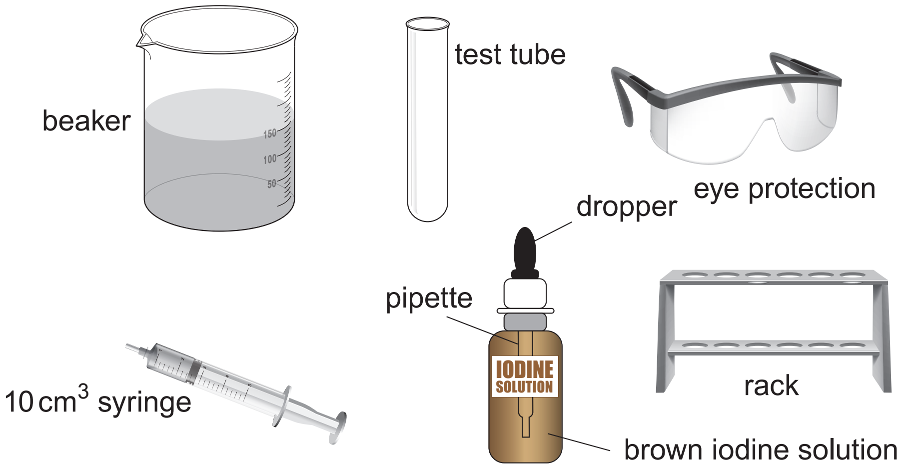

Diagram not drawn to scale

- Using **all** the materials shown in **Image 5** describe how you would:
  - transfer 10 cm^3 of the solution from the beaker to the test tube
  - test it for starch
- State the colour **change** if starch is present. [6 QER]

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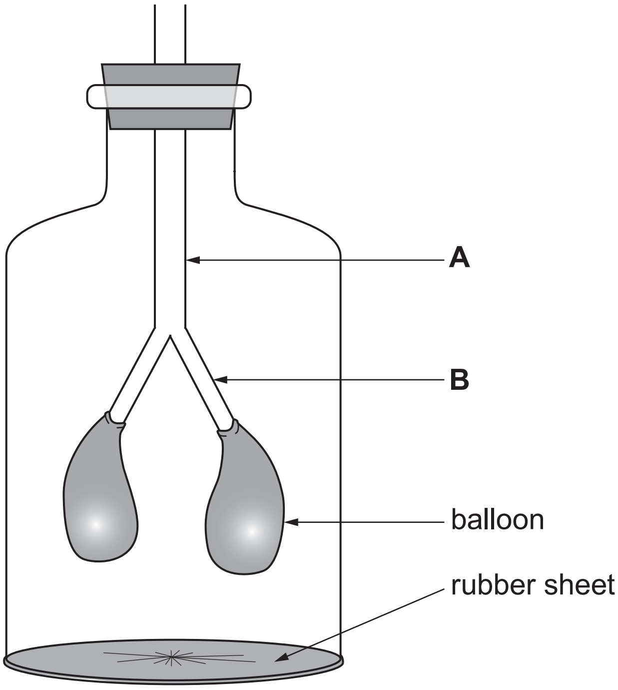
Examiner only

6

Examiner only

6. **Image 6** shows the bell jar model which is being used to demonstrate inspiration.

Image 6



(a) (i) Parts **A** and **B** in **Image 6** represent two parts of the human respiratory system.

Complete the following sentence: [2]

In the bell jar model, part **A** represents the .... and part **B** represents the ....

(ii) As the rubber sheet is pulled down, the air pressure inside the bell jar changes. Explain how the air pressure change in the bell jar causes the balloons to inflate. [2]

....

....

....

....

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(b) The balloons are in an air-tight space in the bell jar. Lungs are sealed in an air-tight space in the chest.

A wound to the chest means the chest is no longer air-tight. As a result, the wounded person cannot fully inflate their lungs.

Explain why a person with a hole in the chest cannot fully inflate their lungs. [1]

....

....

....

....

5

Examiner only

7. **Image 7.1** and **Image 7.2** show examples of two methods of farming hens for egg production.

Image 7.1 – Intensively farmed chickens

Image 7.2 – Free-range chickens

- Intensive farming methods maximise production by controlling the conditions in which farm animals are kept. Inside animal sheds, temperatures may be kept high and each animal is given limited space.
- In 2015, a survey of 2000 people *(OnePoll)* found that 80% of those questioned always or often bought free-range eggs, even though they were more expensive.

Table 7.3 shows egg production in the UK by intensive and free-range methods between 2006 and 2016.

Table 7.3

Year	Egg production (billion)	
	intensive	free-range
2006	4.1	1.9
2008	4.0	2.1
2010	3.8	2.8
2012	3.5	3.4
2014	3.9	3.3
2016	3.8	3.8

(a) Use **Table 7.3** and the information above to answer the following questions.

(i) Calculate the percentage increase in the production of **free-range eggs** between 2006 and 2016. [2]

Increase = ... %

Examiner only

(ii) Give **one** way that intensive farming methods minimise energy loss from farm animals. Explain your answer. [2]

.....

.....

.....

.....

(b) Suggest **two** reasons why many people buy free-range eggs, even though they are more expensive than eggs produced by intensive farming methods. [2]

.....

.....

.....

.....

(c) (i) State **one** possible **pollutant** from intensive farming methods and explain how it could damage the environment. [2]

.....

.....

.....

.....

(ii) Farmers in Wales who plan to develop intensive methods of food production on rural land must first submit their plans to biologists at Natural Resources Wales. Describe the role of the biologists at both the planning stage and when the intensive farm is operating fully. [2]

Planning stage

.....

.....

When operating fully

.....

.....

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END OF PAPER